Subsidy Design and Carbon Pricing: Evidence from India's Coal Allocation*

Abhishek Deshwal†

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Abstract

Economists widely endorse carbon pricing to internalize climate change-related externalities. Yet carbon prices often coexist with energy subsidies. In India's electricity industry, coal is subsidized through administered prices and rationed supply contracts that favor state-owned generators, so the cleanest generators, most of them private, face the highest fuel costs. This paper estimates how the design of this subsidy impacts the effectiveness and welfare gains of carbon pricing. I measure the producer-level distortion the subsidy creates directly, and recover the wholesale electricity supply curve from cost and contract data, exploiting regulation that ties contracted generators' prices to their fuel costs and to specific buyers. Given obstacles to first-best energy pricing, I evaluate a subsidy redesign that keeps the current subsidy's transfers to incumbent generators but gives every generator the same price of coal at the margin. The redesign reduces CO_2 emissions by 14.6 million tonnes a year, more than a \$40 per tonne carbon tax, and increases welfare by \$4.7 billion a year, 1.8 times the gain from the tax. A carbon tax achieves less because the current design lowers costs most for the dirtiest generators, and the tax must first offset this advantage before it changes which generators produce. Findings suggest that in settings where energy subsidies persist, their design can itself serve as climate policy.

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†School of International and Public Affairs, Columbia University. ad3884@columbia.edu

1 Introduction

Reducing greenhouse gas emissions is one of the defining challenges of our time. Standard economic theory prescribes a clear solution: tax emissions at their marginal external cost (Pigou (1932); Nordhaus (1993)). The tax reallocates production to reflect social costs, so that for any level of output, production shifts toward the producers that impose the lowest costs on society. This Pigouvian logic underpins major climate policies worldwide, such as the European Union Emissions Trading System, international carbon offset markets, and border carbon adjustment mechanisms, which together govern incentives for an increasing share of global production.

Recent evidence shows that pre-existing distortions can reduce the efficiency (Borenstein and Bushnell (2022)) and effectiveness of carbon pricing (Preonas (2024); Domínguez-Iino (2024)). In energy markets, among the most widespread of these distortions are subsidies. Global energy subsidies amounted to about $7 trillion in 2022, and often coexist with carbon pricing: in 2022, the European Union and China spent $310 billion and $2 trillion respectively in fossil subsidies (Black et al. (2023)). These subsidies continue to persist because they serve political and economic objectives that make them difficult to reverse: affordable energy, protection for domestic producers, and political support. Therefore, given obstacles to a first-best energy pricing, how these subsidies are designed is of first-order importance for welfare, especially their impact on the efficiency of various environmental policies, such as carbon taxes or cap-and-trade. While existing work estimates the economic impacts of energy subsidies (Davis (2014)), few studies have estimated how they impact the effectiveness of a carbon price.

This paper investigates how the design of India's coal subsidy affects the effectiveness and welfare gains of carbon pricing. India is among the world's fastest-growing sources of CO_2 emissions, with coal accounting for 77.8% of electricity generation and 40% of India's CO_2 emissions, equivalent to 3.5% of global emissions (Ritchie et al. (2020)). Given India's projected economic growth and energy demand, decarbonizing its electricity sector is central to any credible global climate strategy (International Energy Agency (2023)). The question is also immediate: India is introducing a Carbon Credit Trading Scheme that will cover its electricity sector (Bureau of Energy Efficiency (2023)), so a carbon price will soon operate on top of the coal subsidy.

The subsidy operates through the administered pricing and allocation of coal. Coal

India Limited, a state-owned firm, supplies 84% of domestic coal at administered prices of approximately 40% of international benchmarks. Access to this coal is rationed through long-term Fuel Supply Agreements, and allocation rules give priority to public sector generators. Generators with insufficient contracted coal, most of them private, buy the remainder in the domestic markets or through imports, at costs up to 2.5 times higher. Yet private generators are cleaner and more efficient: they burn 10–15% less coal per unit of electricity than public generators built before 2009, which make up the majority of public capacity. As a result, the generators with the lowest private costs are among the most emissions intensive.

This allocation undermines the economic mechanism through which a carbon tax reduces emissions. An emissions tax raises each generator's cost in proportion to its emissions intensity, but when the existing subsidy already protects the dirty generators from the carbon price signal, the tax must first offset that cost advantage before it changes which generators become marginal or are pushed out of production entirely. The strength of this effect is governed by the covariance between the distortion and emissions intensity across generators. When the covariance is negative, the distortion acts like a negative carbon price, clean generators are pushed to the margin, and aggregate emissions are higher, compared with the scenario with no distortionary subsidies. Rather than adjusting the tax to the distortion, this paper evaluates a redesign of the subsidy itself, which sets the distortion to zero so that each generator's cost reflects its true efficiency, while keeping the subsidy transfers to generators intact. How much the distortion reduces the effectiveness of a carbon tax, and how much welfare improvement the redesign would achieve, are empirical questions. The answers depend on three objects: the distortion each generator faces, its covariance with emissions intensity, and which generators are marginal on each day, and therefore which are pushed out when costs change.

I start by measuring the distortion and its relationship with emissions intensity directly. Using detailed plant-level data on coal receipts by source, coal prices, quality, and consumption from 2018 to 2023, I compute each generator's marginal cost of production each month. I then recompute the marginal cost under a uniform price benchmark in which every generator pays the same administered (or subsidized) price per unit of energy in coal. The difference between the two costs is the distortion created by the current subsidy design. It is zero for generators that receive all their coal under supply contracts, and large for generators that source the coal from domestic

markets and imports, most of which are private. I measure a generator's emissions intensity from its heat rate and the carbon content of its coal.

Next, I estimate how generators' position along the supply curve responds to changes in costs, on each day, under various policies. Wholesale prices in Indian electricity markets are heavily regulated: generators with power purchase agreements (PPAs), long-term contracts with large state electricity buyers, can only charge a variable cost that is restricted to their marginal fuel costs and a regulated pass-through markup, and 88% of electricity is traded through these contracts. I construct each generator's fuel costs from its coal prices, coal quality, transportation costs, and fuel efficiency (heat rate), and verify that they line up closely with the variable costs that generators report to the grid operator each day. Two further features of the regulation complete the supply estimation. First, a contracted generator declares itself unavailable when its cost exceeds its regulated price, even when demand is high and the next generator is more expensive; this is why observed production does not follow a standard least cost dispatch, as is often assumed in electricity papers (Hausman (2025); Holland et al. (2026)). Second, generators without contracts bid their marginal cost in a perfectly competitive day-ahead market that meets the residual demand unmet by the contracted capacity. I test the model's main economic mechanism directly using shocks to the price of imported coal, which raise the costs of private generators but not of public, and find that generators differentially change production in response to cost shocks, holding demand fixed. State-level demand is inelastic in the short run because consumers face regulated retail tariffs that change only once a year, and my empirical estimates of the demand elasticity are statistically indistinguishable from zero. The equilibrium on each day is then the lowest cost allocation that meets demand in each state, given generators' available capacity, their regulated prices, and the contracts that tie them to the states they supply. Because costs are observed and prices are set by regulation, the model requires no explicit assumptions about firms' conduct or constraints which is typically the approach in papers modeling generator decisions in electricity markets (Bushnell et al. (2008); Davis and Hausman (2016) ;Preonas (2024); Elliott (forthcoming)). Yet, across more than 200 generators and 1,000 days it explains 85% of the daily variation in generators' output.

With the three empirical objects in hand, I simulate policies that change a single economic primitive: generators' marginal costs. Contracts, regulation and demand are held fixed. The distortion enters the supply curve through costs, so the subsidy

redesign sets each generator's distortion to zero, and a carbon tax raises each generator's cost in proportion to its emissions intensity. I compare the redesign with carbon taxes of $20 and $40 per tonne; a $40 tax is significant and roughly doubles the average marginal cost of electricity production. Each counterfactual changes generator costs and therefore, their relative position along the supply curve: some generators become inframarginal, others are pushed out of production, and contracted generators whose costs fall below their regulated price become available to produce. Solving for the new equilibrium under each policy gives the resulting changes in production, emissions, and costs across generators, and comparing them yields the economic costs of the subsidy design.

I find that the distortion is concentrated among the cleanest generators under the current subsidy design. Public generators, which receive most of their coal under subsidized prices, face a distortion of 5 percent of their true marginal cost on average, while private generators face a distortion of 43 percent. As a result, the cleanest decile of generating capacity, 79 percent of it private, faces a marginal cost increase of INR 1.71 per kWh, nearly two-thirds of its true marginal cost, while the dirtiest decile, 86 percent of it public, faces only INR 0.33 per kWh. This gap is large relative to the carbon price signal. Because the dirtiest generators emit about 50 percent more CO_2 per unit of electricity than the cleanest, a $20 tax offsets only half the gap the distortion creates, and a $40 tax does little more than cancel it.

My main result is that redesigning the subsidy reduces emissions by more than a moderate carbon tax, at a much larger welfare gain. The redesign keeps each generator's existing coal contracts and compensates generators for the coal they buy at market prices, so that every generator faces the same price of coal at the margin. It reduces CO_2 emissions by 14.6 million tonnes a year, more than a $40 tax achieves under the current design (12.8 million tonnes), with a total welfare gain of $4.7 billion a year that is 1.8 times that of the tax. I estimate larger impacts from redesign because the distortion changes which generators produce at the margin at different levels of carbon pricing. Under status quo subsidy, the marginal generators are progressively cleaner as demand rises, the opposite of an efficient allocation in which the cleanest generators would be inframarginal and dirtier generators called on only at high levels of demand. A carbon tax is unable to reverse this sufficiently. Under a $20 tax, the generators pushed out of production are on average only 2 percent dirtier than the fleet, and on a quarter of days they are cleaner. On the other hand, redesigning the

subsidy moves production towards the efficient allocation: the generators it pushes out are 6 percent dirtier than the fleet and never cleaner, and production shifts from public generators to cleaner private ones, with private generation rising by 24 percent. The climate benefit alone therefore exceeds that of the $40 tax, and because production moves to generators that burn less coal and are cheaper to supply, it also saves $1.9 billion a year in fuel and transport costs. The tax on status quo subsidy realizes very little of these savings.

Finally, because the status quo subsidy absorbs much of the carbon price signal, it also raises the carbon price needed to reach any emissions target. With the subsidy redesigned, a $20 tax reduces emissions by 21.8 million tonnes a year, a reduction that requires an $80 tax under the status quo; a $40 tax reduces emissions by 26.4 million tonnes, which requires a $100 tax under the status quo. Redesigning the subsidy therefore lowers the carbon price required to meet these targets by a factor of two and a half to four. For a country introducing carbon pricing on top of existing energy subsidies, the design of those subsidies determines the efficiency of the carbon price, its effectiveness, and since carbon taxes introduce costs, how costly it would be to achieve a given reduction in emissions.

These findings have a direct implication for environmental policy design: when significant obstacles to removing existing subsidies exist, producer incentives can be aligned by changing the subsidy each producer receives at the margin. I show in the paper that an efficient subsidy design makes the subsidy decrease with emissions intensity. This is precisely the mechanism that makes cap-and-trade the least-cost way of reducing emissions: a common permit price at the margin moves production toward the producers that can supply it at the lowest social cost. The uniform design I evaluate is a simpler version of this principle, and in this setting it closely tracks the shape of the supply of the efficient one because true private and external costs are positively correlated across thermal generators. Once the distortion is removed, the marginal emissions curve closely tracks the efficient one. The redesign also relies on institutions that already exist, administered coal prices and long-term supply contracts, rather than new regulatory or enforcement capacity (Greenstone and Jack (2015)). Correcting the design of existing subsidies is thus a practical complement to carbon pricing in precisely the economies where carbon pricing is being introduced.

In terms of related literature, my paper makes three main contributions. First, I contribute to the literature on misallocation in developing countries, which has docu-

mented massive efficiency losses (Hsieh and Klenow (2009); Restuccia and Rogerson (2017)), and has identified state enterprise favoritism as an important driver (Dollar and Wei (2007)). This literature measures the costs of misallocation in terms of output and productivity. I show that misallocation also has environmental costs, and that it changes the impacts of environmental policy: when a distortion protects producers that are both less efficient and more emissions intensive, it raises aggregate emissions and absorbs the price signal of a carbon tax. Correcting the distortion is then itself an environmental policy. Redesigning India's coal subsidy reduces emissions by more than a $40 carbon tax, at 1.8 times the welfare gain. In this respect, the paper also relates to work emphasizing that institutional constraints shape both environmental quality (Greenstone and Jack (2015); Berkouwer and Dean (2022); Baylis et al. (2023)) and the success of environmental policy in developing countries (Duflo et al. (2013); Greenstone and Hanna (2014); Oliva (2015); Duflo et al. (2018); Jayachandran (2022)).

Second, I contribute to the literature on second-best environmental policy (Lipsey and Lancaster (1956); Bovenberg and Goulder (1996)). Recent empirical work shows that market power can mute the pass-through of a carbon price and reduce its effectiveness (Xia and Zhang (2023); Preonas (2024); Domínguez-Iino (2024)). I study a new distortion, a subsidy that remains in place, and ask how its design changes the effectiveness of carbon pricing. I show that the answer depends on a single statistic, the covariance between the distortion and emissions intensity across producers, and that it can be measured directly. In India, the status quo subsidy design raises the carbon price needed to achieve a given reduction in emissions by a factor of two and a half to four. Unlike market power, which arises from market structure, the distortion I study is created by policy, and can therefore be corrected through policy without removing the subsidy itself.

Third, I contribute to a rich literature studying supply-side distortions in wholesale electricity markets by identifying a novel source of inefficiency and connecting these distortions to environmental policy design. Previous work on electricity market inefficiencies focuses on cost-of-service regulation and market structure (Fabrizio et al. (2007); Davis and Wolfram (2012); Cicala (2022)), while recent work on India examines these same regulatory distortions (Ryan (2021a); Jha et al. (2021)). I extend this literature by identifying upstream market distortions, specifically administrative coal allocation, as a major source of supply-side misallocation, documenting $1.9 billion in

annual efficiency losses from fuel market distortions alone. Unlike previous work that focuses on production costs, blackouts, and reliability impacts, I am the first to connect electricity market distortions to environmental policy effectiveness, showing that removing upstream input distortions can achieve large emission reductions, and deliver larger welfare gains than direct environmental regulation. This finding suggests that the focus of the electricity market literature on downstream regulatory reforms may miss significant efficiency gains available from upstream market reforms, with broader relevance as distortions often transmit along the supply chain with aggregate implications (Acemoglu et al. (2012); Barrot and Sauvagnat (2016)).

The rest of the paper proceeds as follows. Section 2 explains the theoretical framework. Section 3 provides institutional background on India's power sector and details the evolution of coal allocation policies and describes the data. Section 4 provides empirical evidence on coal market distortions. Section 5 describes the model and empirical strategy. Section 6 presents my main results, analyzing both the environmental costs of coal market distortions, and quantifying gains under counterfactual environmental policies. Section 7 concludes with policy implications.

2 Theoretical Framework

This section sets out a simple framework to analyze when a supply-side distortion (eg. coal subsidies) blunts carbon pricing and why removing it reallocates production in ways a carbon tax cannot. The answer lies in the joint distribution of the distortion and the external cost across producers: when the distortion raises the private costs of the producers with the lowest external costs, a carbon tax must first offset the distortion before it can change producers' positions along the supply curve. Each mechanism maps to an object I measure in Section 6: the correlation between the distortion and emissions intensity, the shape of the marginal emissions curve, and the emissions intensity of the generators that a policy pushes out of production.

The framework applies to any market in which, absent the distortion, true private costs and external costs are positively correlated across producers, and a pre-existing distortion raises the private costs of the efficient, clean producers relative to the inefficient, dirty ones, so that the distortion is negatively correlated with the external cost. Preferential access to subsidized inputs, credit or contracts for state-owned or politically connected incumbents, which a large literature documents across developing-country industries (Hsieh and Klenow, 2009; Restuccia and Rogerson, 2017), gener-

ates exactly this joint distribution whenever the beneficiaries are also less efficient and more polluting. India's electricity industry, in which state-owned generators receive coal at subsidized administered prices and more efficient private generators buy coal at market prices, is one such market.

2.1 Illustrative Framework

Figure 1 illustrates the mechanism. Each step of the supply curve is one producer, and producers are ordered by private marginal cost. Demand is met by the producers with the lowest private costs. The producer that serves the last unit of demand is the marginal producer; producers to its left are inframarginal and produce at capacity; producers to its right are pushed out of production. Panel (a) depicts the benchmark without the distortion. Producers face their marginal cost c^u (solid line), and the dashed line adds the marginal external cost from their emissions, τe, giving their social marginal cost. Because true marginal cost and emissions intensity are positively correlated, ordering producers by their marginal private cost also orders them *roughly* by social marginal cost. A carbon tax shifts the supply curve from the solid toward the dashed line and moves production further away from generators with high social costs.

Panel (b) shows the supply curve with the distortion. Producers face private marginal cost $c^u + d$, where the distortion d raises the costs of the cleanest producers.[1] The supply curve is still upward sloping but producers' positions along it no longer reflect their *true* social marginal costs.

The dash-dotted line adds a carbon tax to the distorted private costs. It is non-monotone: some of the producers with the lowest private costs are among the dirtiest, so they sit at the bottom of the supply curve but near the top of any ranking by social cost.

Panels (c) and (d) plot the marginal emissions curves for the two supply curves. Without the distortion, the cleanest producers are inframarginal at every level of demand, and the producers that serve additional demand are progressively dirtier: the marginal emissions curve slopes upward. With the distortion, clean producers move up the supply curve and dirty producers move down it. Dirty producers be-

[1] Only relative costs determine producers' positions along the supply curve, so a subsidy to the dirtiest producers is equivalent to a cost premium on the others. I write the distortion as this premium, $d \geq 0$, which is the quantity measured for India's coal-fired generators in Section 5.2.

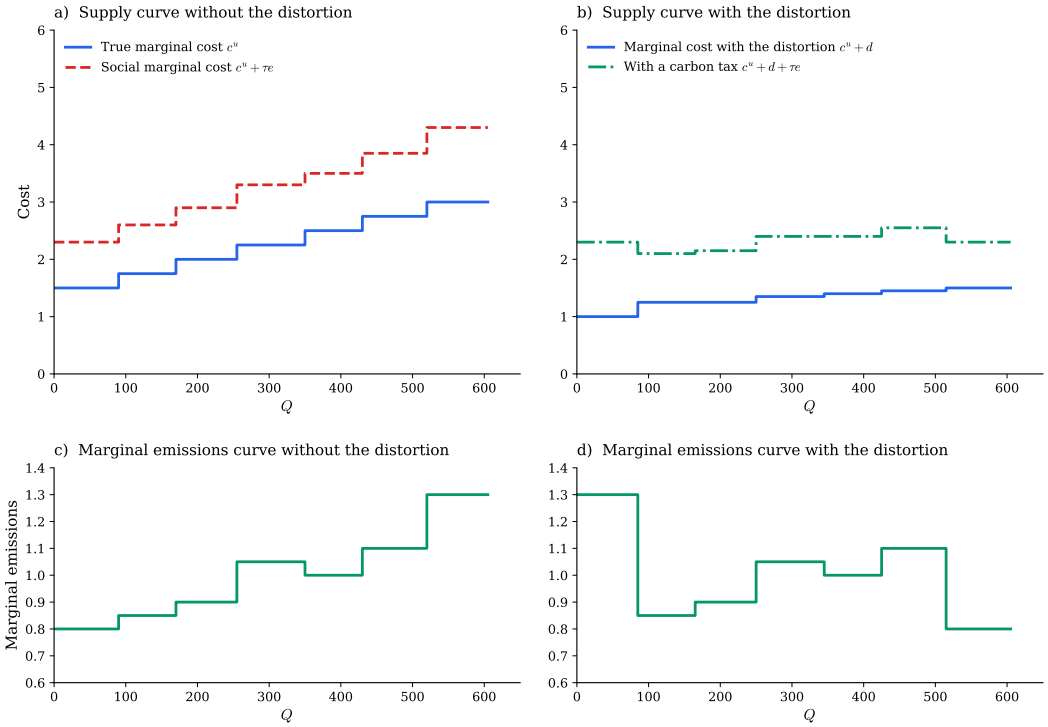
come inframarginal, while clean producers are held at the margin or pushed out of production, and the marginal emissions curve becomes flat or slopes downward. The shape of the marginal emissions curve is therefore a direct test of whether production is allocated to reflect *true* social costs. Section 6.3 estimates these curves for India's wholesale electricity markets.

This has a direct implication for carbon pricing. A tax at rate τ adds τe_i to each producer's private marginal cost. It improves the allocation of production only if it changes producers' positions along the supply curve, moving production away from those with high social costs. When the distortion favors dirty producers, a moderate tax raises the costs of all producers without changing their positions relative to one another: the marginal producer, and therefore how much each producer supplies, changes little, and so do emissions.[2]

Removing the distortion reverses this. Setting $d_i = 0$ returns each producer's private marginal cost to its true private cost, and the supply curve realigns with the social cost ordering of panel (a): production moves toward cleaner and more efficient producers, without a carbon tax. I formalize these results below.

[2]This logic is exact when demand is inelastic, which I assume in the short run. With elastic demand, a carbon tax also reduces aggregate output, but the reallocation channel, through which carbon pricing changes which producers supply the market, remains weak when the distortion favors dirty producers.

Figure 1. Marginal cost and social marginal cost curves with and without coal subsidies



In panels (a) and (b), each step of the supply curve is one producer, with width equal to its capacity, and producers are ordered by private marginal cost. Panel (a): without the distortion, private marginal cost equals true marginal cost c^u (solid line); the dashed line adds the damage from emissions, τe, giving social marginal cost. Because true marginal cost and emissions intensity are positively correlated, producers are ordered by social marginal cost as well. Panel (b): with the distortion, private marginal cost is $c^u + d$, where the distortion raises the costs of the cleanest producers relative to the dirtiest (solid line); the dash-dotted line adds a carbon tax, τe, to these costs and is non-monotone, so producers' positions along the supply curve no longer reflect their social marginal costs. Panels (c) and (d): the emissions intensity of the marginal producer, the producer that serves the last unit of demand, at each level of demand Q, for the supply curves in panels (a) and (b). Without the distortion, the cleanest producers are inframarginal and the marginal emissions curve slopes upward; with the distortion, dirty producers are inframarginal while clean producers are held at the margin, and the marginal emissions curve slopes downward.

2.2 Model Setup

Consider producers i that differ in true marginal cost c_i^u and emissions intensity e_i (tonnes of CO_2 per unit of output). Without the distortion, more efficient producers use fewer inputs and emit less, so $\text{Cov}(c_i^u, e_i) > 0$. The distortion $d_i \geq 0$ raises producer i's private marginal cost to $c_i = c_i^u + d_i$. When the distortion is large and concentrated among clean producers, $\text{Cov}(d_i, e_i) < 0$, the correlation between

private costs and emissions weakens or reverses, $\text{Cov}(c_i, e_i) < 0$, and the distortion raises aggregate emissions. Figure 2 shows the joint distribution of costs and emissions intensities that underlies the two supply curves in Figure 1. The left panels correspond to the benchmark of panel (a), in which efficient producers have both lower costs and lower emissions; the right panels correspond to the distorted supply curve of panel (b), in which the distortion reverses this pattern. Panel A of Figure 2 shows the distribution without a carbon tax. Each bubble is a producer, with its private marginal cost on the horizontal axis, its emissions intensity on the vertical axis, and its size proportional to its capacity utilization. The red circles highlight the region in which dirty, state-owned producers operate. Section 6.2 shows the empirical counterpart of this distribution.

Under a carbon tax τ, producer i faces private marginal cost $c_i^u + d_i + \tau e_i$, and inelastic demand D is met by producers in ascending order of this cost. Replicating the undistorted allocation requires producers' positions along the supply curve under $c_i^u + d_i + \tau e_i$ to match their positions under $c_i^u + \tau e_i$, which requires an additional, offsetting tax τ^*. The magnitude of this offsetting tax measures the efficiency loss from the distortion relative to the undistorted benchmark. It also measures what removing the distortion achieves: removing the distortion reallocates production as a tax of τ^* would, reducing both private costs and emissions.

2.3 Analysis

Static analysis of carbon taxation with supply-side distortions. Consider a social planner who minimizes social costs, private costs plus the damage from emissions, while meeting inelastic demand D. Carbon tax revenue is returned to consumers lump sum and does not enter the welfare calculation. The planner's first-best problem is:

$$\min_{q_i} \sum_i (c_i^u + \tau_s e_i)\, q_i \quad \text{subject to} \quad \sum_i q_i = D, \quad 0 \le q_i \le \bar{q}_i, \tag{1}$$

where q_i is the output of producer i, τ_s is the social cost of carbon, and $\bar{q}_i$ is capacity. The solution allocates production to producers in ascending order of social marginal cost, $c_i^u + \tau_s e_i$. Without the distortion, the market implements this allocation: a carbon tax of τ_s orders producers along the supply curve by $c_i^u + \tau_s e_i$, the first best.

With the distortion, producers face $c_i^u + d_i$, and under a carbon tax of τ_s they are

ordered by $c_i^u + d_i + \tau_s e_i$. The distortion breaks the efficient ordering, and τ_s is no longer the first-best policy. A planner who cannot remove the distortion can instead choose a tax $\tau = \tau_s + \tau'$, where τ' is a corrective component, so that producers are ordered by

$$c_i^u + d_i + (\tau_s + \tau')\, e_i.$$

This ordering matches the first-best ordering by $c_i^u + \tau_s e_i$ if $d_i + \tau' e_i$ is the same for all producers, since a common shift in every producer's cost does not change their positions along the supply curve. Since the planner cannot eliminate d_i, she uses the corrective tax $\tau' \cdot e_i$ as a second-best instrument to offset the distortion.

Optimal Corrective Tax.

The planner chooses τ' that minimizes the net distortion term $(d_i + \tau' e_i)$ across all plants:

$$\min_{\tau', a} \sum_{i=1}^{N} (d_i + \tau' e_i - a)^2,$$

where a is a common shift that does not affect the allocation. The first-order conditions give

$$\tau^* = -\frac{\mathrm{Cov}(d_i, e_i)}{\mathrm{Var}(e_i)}.$$

The optimal tax is $\tau_s + \tau^*$.

Economic Interpretation: What Does τ^* Measure?

τ^* is the additional carbon tax needed to offset the distortion. It is positive when the distortion falls on clean producers, $\mathrm{Cov}(d_i, e_i) < 0$, and large when the distortion is strongly correlated with emissions relative to how much emissions intensity varies across producers. Two implications follow, and both hold in any market with the joint distribution described above.

1. A measure of misallocation. $|\tau^*|$ increases with the covariance between the distortion and emissions intensity and decreases with the dispersion of emissions intensity. In markets in which producers use similar technologies, so that emissions intensity varies little, even a moderate distortion requires a large offsetting tax. This is the case among coal-fired generators, and more generally in industries in which producers differ mainly in efficiency rather than in fuel or technology.

2. A threshold for carbon tax effectiveness. Consider a carbon tax τ imposed

on the distorted market. Three cases emerge:

- If $\tau < \tau^*$: the tax only partly offsets the distortion and reallocates little production.
- If $\tau = \tau^*$: the tax offsets the distortion and reallocates production as removing the distortion would, but does not price emissions.
- If $\tau = \tau_s + \tau^*$: the tax achieves the first best.

When τ^* is large, a carbon tax at the social cost of carbon falls well short of the first best, and moderate taxes are largely absorbed by the distortion. Removing the distortion ($d_i = 0$, hence $\tau^* = 0$) achieves the reallocation of a tax of τ^* without any tax, and a tax of τ_s then achieves the first best. Figure 3 illustrates this threshold in the space of producer costs and emissions. Each panel shows the distorted market from the right panels of Figure 2 under a carbon tax that increases from left to right. As the tax rises toward $\tau_s + \tau^*$, the threshold above which producers are pushed out of production rotates and progressively excludes the dirtiest producers, moving the allocation toward the efficient ordering of the left panels of Figure 2, and equivalently toward the supply curve of panel (a) in Figure 1. Section 6.6 compares the distortion with the carbon tax directly: between the cleanest and dirtiest tenths of India's coal and lignite capacity, a $20 tax offsets about half of the cost gap the distortion creates, and a $40 tax roughly cancels it.

Optimal subsidy design. The corrective tax has a direct implication for the design of subsidies. Suppose political constraints rule out removing input subsidies, for instance because the government wants to keep output prices low, but allow it to change how subsidies are allocated across producers. The condition that $d_i + \tau^* e_i$ be constant across producers implies that, for a given subsidy budget, the distortion should satisfy

$$d_i \approx a - \tau^* e_i.$$

Subsidies should be *decreasing* in emissions intensity: the cleanest producers should receive the largest subsidies, and the dirtiest the smallest. Allocations that favor incumbents do the opposite whenever the incumbents are the dirtiest producers. In India, generators without access to coal at the notified price, which are disproportionately the cleanest, receive no subsidy.

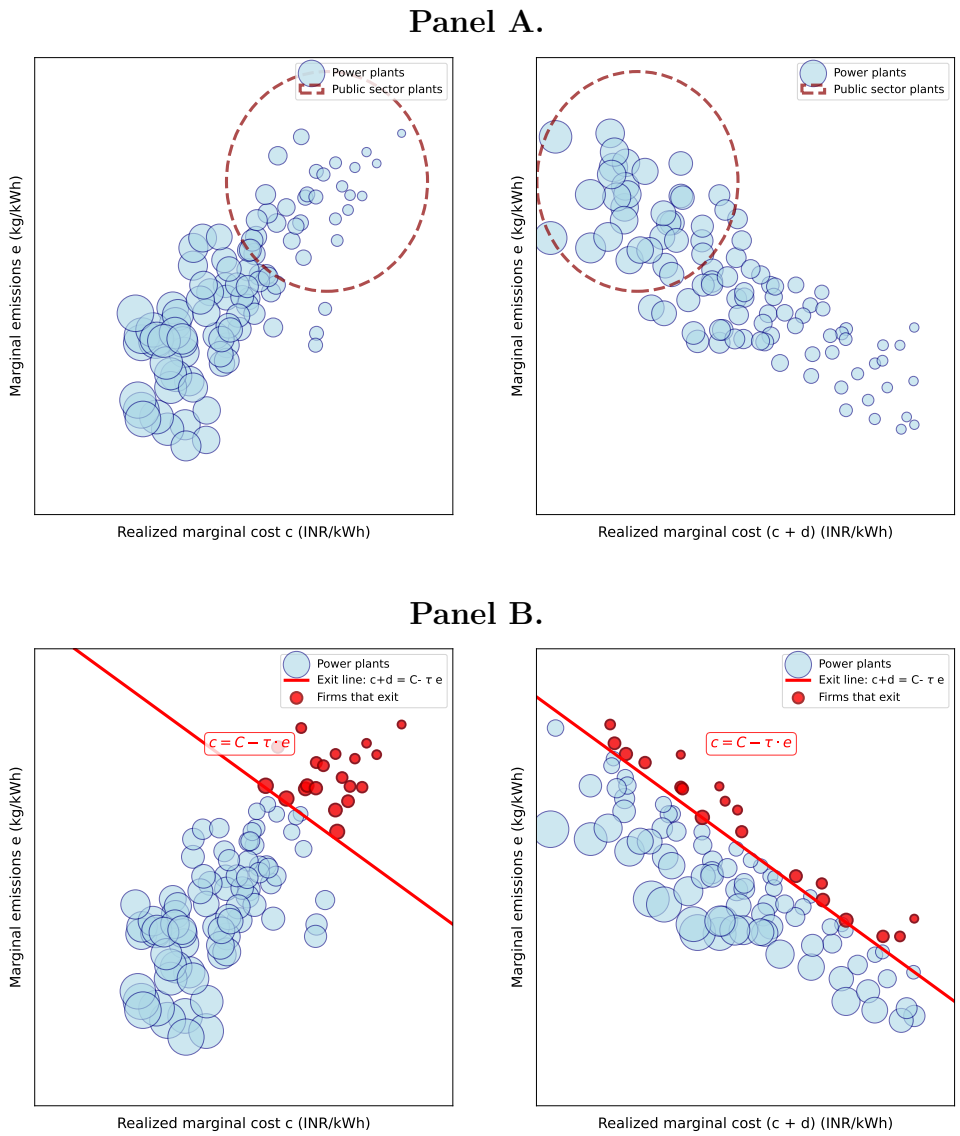
A uniform subsidy, the same input price for every producer, is the neutral benchmark:

it sets $d_i = 0$ and removes the misallocation without reducing the subsidy budget, although it does not price emissions. The compensatory tariff I simulate in Section 6.6 implements this benchmark. Generators continue to receive coal under their fuel supply agreements at the notified price, and the government reimburses the difference between the market and notified price on all coal bought on the market, so that every generator faces the notified price per kcal at the margin. A subsidy that decreases in emissions intensity would go further, combining the reallocation of removing the distortion with part of the effect of a carbon tax.

Medium to long-run dynamics. The same logic extends to the medium and long run, when demand is elastic and producers can exit. Let $\bar{C}$ be the break-even revenue: producers whose costs exceed it cannot operate profitably, and it equals the market price, the private marginal cost of the marginal producer. Producer i exits when $c_i^u + d_i + \tau e_i > \bar{C}$.

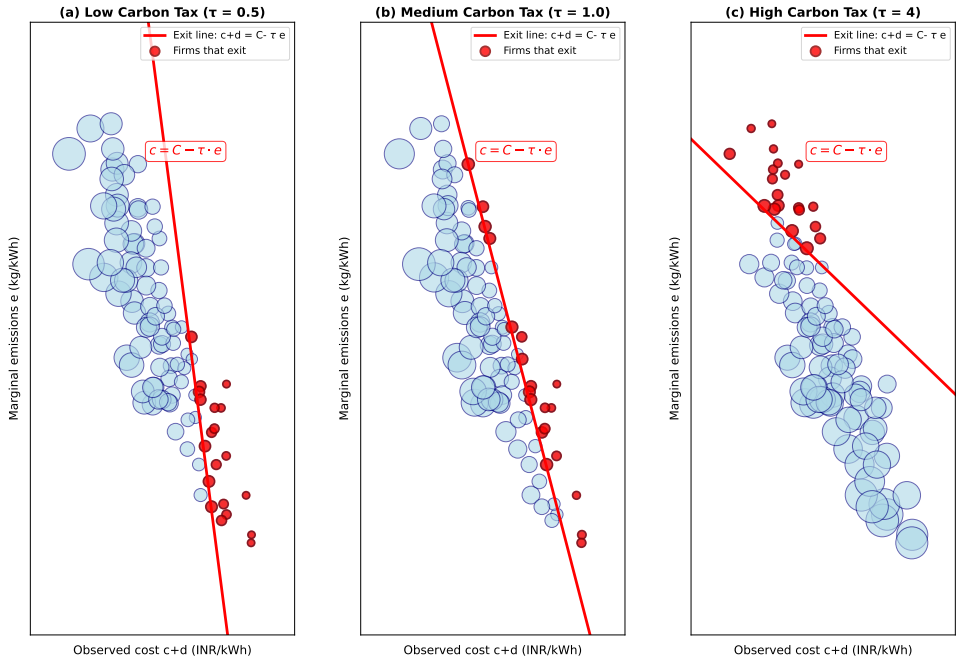
Panel B of Figure 2 illustrates the mechanism. The red line shows the exit threshold, $c = \bar{C} - \tau e$; producers above it exit. Without the distortion (left panel), the positive correlation between costs and emissions ensures that a carbon tax targets the right producers: those that are both dirty and inefficient lie in the upper-right region and exit first. With the distortion (right panel), $\text{Cov}(d_i, e_i) < 0$: clean producers face high private costs because of the distortion, while dirty producers are shielded from it. The exit threshold cuts through both clean and dirty producers, and the carbon tax pushes out clean producers along with dirty ones. Among the producers that remain, clean and dirty producers both operate at high capacity utilization, so the tax does little to reallocate production toward cleaner producers. As the tax rises, the distortion is progressively offset, and once it reaches $\tau_s + \tau^*$ producers' positions along the supply curve replicate the social cost ordering of panel (a) in Figure 1. Figure 3 traces this progression. In industries in which carbon pricing is expected to drive the exit of the dirtiest incumbents, a distortion of this form can instead drive the exit of the cleanest entrants.

Figure 2. Carbon pricing with distortion



Note: c denotes generators' marginal operating cost in INR per kWh, and e denotes their emissions rate (kg CO_2 per kWh). Each bubble represents a power plant, with bubble size indicating the plant's capacity factor (fraction of available capacity utilized for generation). In panel A, the dotted red circles indicate the regions in which inefficient public generators operate. In panel B, the red line shows the exit threshold under a carbon tax τ, where generators with total cost $c+\tau e$ exceeding the break-even revenue C are forced to exit the market (equivalently, $c > C - \tau e$). The left figures in both panels A and B show an undistorted economy where efficient generators tend to have lower emissions (positive correlation between c and e). The right figures in both panels A and B show a distorted economy where input subsidies create a negative correlation between operating costs and emissions, such that cleaner generators face higher costs due to limited access to subsidized inputs. Red dots indicate generators that exit under the carbon pricing policy.

Figure 3. Targeting under varying carbon taxes



Note: c denotes generators' marginal operating cost in INR per kWh, and e denotes their emissions rate (kg CO_2 per kWh). Each bubble represents a power plant, with bubble size indicating the plant's capacity factor (fraction of available capacity utilized for generation). All panels show the distorted economy from the right panels of Figure 2, corresponding to the subsidized supply curve of panel (b) in Figure 1. The red line in each panel shows the exit threshold under carbon tax τ, where generators with total cost $c + \tau e$ exceeding break-even revenue $\bar{C}$ are forced to exit. τ increases from left to right. The rightmost panel, where $\tau \approx \tau_{social} + \tau^*$, restores the efficient dispatch ordering of panel (a) in Figure 1. Red dots indicate generators that exit under the carbon pricing policy.

3 Context and Data

3.1 Dominance of Coal in India's Electricity Generation

Coal remains the backbone of India's electricity generation, accounting for 77.8 percent of total monitored generation from 2018 to 2023. This share has stayed remarkably stable despite significant renewable capacity additions and global pressure to transition away from fossil fuels. The growth of non-hydro renewables from 5.9 to 8.9 percent has largely come at the expense of gas generation rather than coal, with gas falling to just 1.8 percent of generation despite comprising 6 percent of installed capacity (Figure 4). Coal's sustained dominance reflects not just path dependence but the reality that renewable additions have primarily accommodated growing electricity demand rather than displacing existing thermal capacity.

India's coal dependence stands in sharp contrast to trends in developed economies. While the United States has seen both the share and absolute level of coal generation decline substantially since 2010 (Appendix figure A1), India follows a trajectory similar to China's earlier industrialization, with coal generation continuing to grow in absolute terms. This pattern shows little sign of reversing. The International Energy Agency (2023) projects that India will drive the world's largest increase in energy demand over the next three decades, fueled by a 74 percent rise in urban population, tripling per capita incomes, and rapid industrial growth. Even with ambitious solar deployment, the IEA still forecasts a 10 percent increase in coal demand by 2050. Given these demand pressures, existing infrastructure investments, and the intermittency of renewables, coal will likely anchor India's power system for decades to come. The question then is not whether India will continue using coal, but how to use it more efficiently while reducing environmental costs.

Figure 4. Electricty generation by source in India

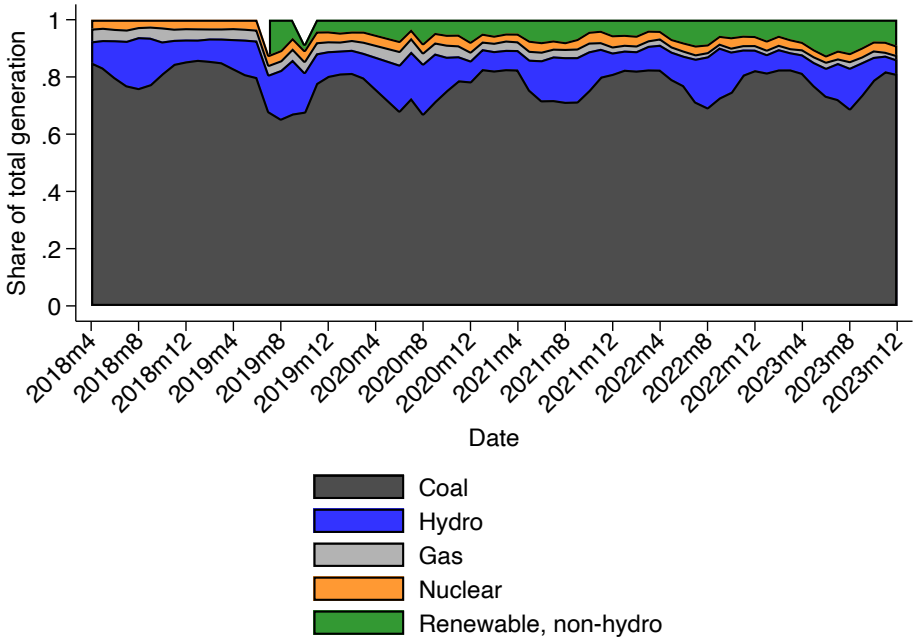


Figure plots monthly shares of total electricity generation in India by source from 2018 to 2023. The y-axis shows the fraction of total generation contributed by each source: coal (dark gray), hydro (blue), gas (light gray), nuclear (orange), and non-hydro renewables (green). Non-hydro renewable generation data are included from April 2019 onwards, when daily frequency reporting for these sources began. Source: Central Electricity Authority (2018-2023a)

3.2 Private sector participation in India's electricity generation industry

After independence in 1947, the Indian power sector was mostly publicly owned and run for more than 40 years. A sweeping economic liberalization in the early 1990s opened power generation to private firms, and the Indian Government solicited investment, including from foreign companies. For the power sector specifically, this period of liberalization was a failure. The deregulation of entry in generation reduced state control over generation, but retained the state's primary role in the transmission and distribution of electricity. As deregulation was not paired with any deeper structural reform, potential private entrants still faced monopsony buyers, the State Electricity Boards, in every state, and were therefore reluctant to invest (Mathavan (2008)).

Figure 5 shows the thermal generation capacity in the Indian electricity Industry from 1990 to 2019, and capacity by ownership (state and central governments together comprise public, and private). The privately-owned share of capacity is nearly zero (< 1 GW) until 2001, and is slow-growing thereafter.

Figure 5. Growth in installed thermal capacity by ownership

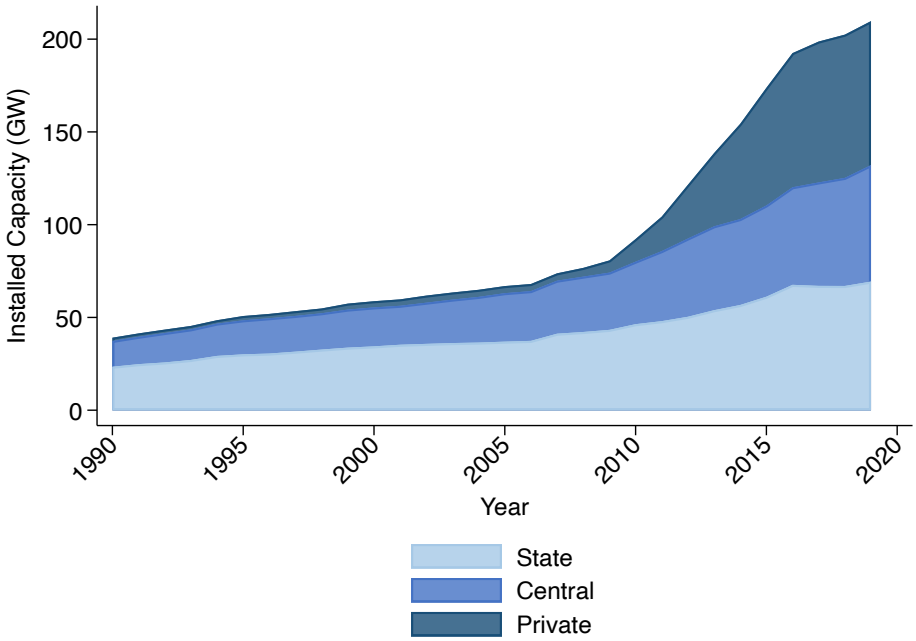


Figure plots the evolution of total installed thermal electricity generation capacity in India by ownership type from 1990 to 2019. The y-axis shows cumulative installed capacity in gigawatts (GW). Stacked areas represent cumulative capacity (in GW) for state utilities (light blue), central government (medium blue), and private sector (dark blue). Each area's height at any given year indicates the total capacity owned by that sector, while the overall height of the stacked areas shows total installed capacity across all ownership types. Source: Central Electricity Authority, India

The failure of this generation of private projects and power rationing built up pressure for deeper reforms. The Electricity Act of 2003 undertook structural reforms that recognized the natural monopoly nature of much of the power sector (Kumar and Chatterjee (2012)). Under these laws, the State Electricity Boards were separated into component parts, for generation, transmission, and distribution. Independent regulators were established to rule on power contracts and tariffs. Markets for power, though initially a small share of the sector, provided an outside option for private entrants to sell power (Ryan (2021b)). The National Tariff Policy (2006), issued under Section 63 of the Electricity Act (2003), mandated that state procurement of power must be done through a competitive bidding process (Ministry of Power (2006)). The auction results are formalized in a power purchase agreement (PPA), a contract between the supplier and procuring state utility or utilities that is written at the price set by the auction and reviewed and approved by the electricity regulator.

Several features of this procurement process made the industry attractive to private power sellers: the existence of Electricity Regulatory Commissions to approve contracts and arbitrate disputes; the principle of contracts being revealed at auction, which may lower contract prices and increase the transparency of price setting; the deference of the institutions now involved to the market process of setting prices; the large amount of power they are supplying, which may make them difficult to replace in the short term; and finally, the fact that the buy side consists of consortia of state bidders for large projects, which diversifies the political risk faced by sellers (Ryan (2020)).

There was a large wave of private investment in power in response to the deeper structural reforms in the electricity sector and under the new bidding rules. The rules took effect in 2006 but the new projects built under these rules typically had a 4-5 year lag, meaning that they came online in 2010. Returning to Figure 5, we see a rapid increase in private generation capacity during this period, as shown by the blue curve, such that by 2019 private generation capacity was over 36 percent of the total capacity, greater than that owned by either the states or the Central Government. This dramatic transformation positioned private generators as major players in India's electricity sector, with many demonstrating superior operational efficiency compared to their public counterparts. Till the final date of this analysis, private sector continues to contribute 36 percent of the total capacity.

3.3 Coal Market Distortions in India's Power Sector

India's power sector is characterized by significant institutional distortions in coal markets that create systematic inefficiencies that distort both operational decisions of power plants and aggregate environmental outcomes. At the core of these distortions lies Coal India Limited's (CIL) near monopoly over coal production and distribution, coupled with regulatory structure that systematically favor public generators over private ones. These institutional arrangements misallocate resources across generators and undermine aggregate environmental performance by reducing the utilization of more efficient private plants while maintaining higher dispatch rates for older, less efficient public generators.

CIL produces 84 percent of India's domestic coal supply, with the remainder from Singareni Collieries Company Limited (SCCL) and captive mines. While CIL holds monopoly power, it doesn't exercise monopoly pricing. Instead, coal is primarily sold

through notified prices, which are administratively determined, grade-specific rates that implicitly carry subsidies. These prices are set to balance CIL's profitability with the government's objective of keeping coal prices low for end-users. Only 10-20 percent of total production is sold through e-auctions, where the premium prices reflect coal's scarcity value Ministry of Coal (2014). Coal typically accounts for 60-70 percent of variable generation costs, making fuel access the primary determinant of plant competitiveness.

The real market for coal in India is essentially access to coal contracts or fuel supply agreements (FSAs) that specify supply guarantees at notified prices, but with explicit priority given to the power sector and further preferential treatment to public sector users. Those lacking secure access to domestic coal must rely on more expensive imports, creating a two-tier market where producer-level economic outcomes depend on administrative fuel access rather than efficiency.

3.3.1 Evolution of coal allocation policies

The institutional framework governing coal allocation evolved through successive policy revisions that systematically disadvantaged private entrants during the period of private sector expansion:

Pre-2007 Discretionary Regime: Coal allocation occurred through the Standing Linkage Committee (Long Term) [SLC(LT)], a body where public sector interests dominated—notably, the only consumer representative was the chairman of NTPC, India's largest public power generator (Sehgal and Tongia (2020)). These allocations, particularly through Memorandum of Understanding (MoU) based arrangements, offered generous terms including high supply guarantees and favorable pricing. Most beneficiaries were public sector plants. The discretionary nature of these allocations faced legal challenges from private generators who were kept out, leading to Supreme Court intervention that mandated a more transparent, rule-based distribution system.

2007 New Coal Distribution Policy (NCDP): The NCDP introduced a structured framework with sector-wise priorities. While improving transparency, the policy focused primarily on ensuring coal availability to the power sector rather than enhancing distribution efficiency. Under NCDP, CIL established two main supply routes: long-term fixed price contracts through Letter of Assurance (LoA) leading to Fuel Supply Agreements (FSAs), and short-term price supply through domestic spot markets or e-auctions (Ministry of Coal (2007)). Private generators, being newer

entrants, predominantly fell under this regime with its stricter terms compared to the generous MoU arrangements that existing public sector plants had secured.

During the NCDP transition, CIL signed standardized FSAs with existing customers, promising 100 percent supply to the power sector; new projects, predominantly private, fell under the LoA-FSA mechanism. However, combined commitments under FSAs and LoAs far exceeded CIL's production capabilities.

2010-2013 Coal Supply Crisis: By 2010-11, as new generators came online, most of them private, CIL struggled to meet the growing demand for coal from these generators. Recognizing this supply constraint, CIL stopped issuing new LoAs. This particularly affected private generators who were still in various stages of project development. A 2012 Presidential directive initially mandated CIL to supply 100 percent coal for 78 GW of upcoming capacity under the 12th Plan (2012-17). However, facing supply constraints, a 2013 revision reduced this to 60 GW, with 11 GW receiving tapering linkages—temporary supply arrangements that decrease as plants' captive mines become operational.[3][4].

Following the revised directive, the Ministry of Coal introduced differential trigger values (minimum supply guarantees)—90 percent for pre-NCDP consumers (mostly public) and 75 percent for new plants commissioned between 2009-2017 (predominantly private)[5]. These trigger values determine penalty thresholds for supply shortfalls, effectively creating different consumer classes based on contracting period. CIL linked contracted coal supply to long-term PPAs, prioritizing existing plants with firm commitments. While many private plants failed to convert their LoAs to FSAs due to strict milestone requirements, public sector plants faced fewer such hurdles. CIL continued to supply to meet demand with lower trigger values and those without firm FSAs, mostly private generators, were forced to rely more on imports.

Import-based plants and Policy-Induced Cost Shocks: Concurrent with domestic coal allocation challenges, private generators pursuing import-based strategies faced their own policy-induced setbacks that further undermined efficient generation. Several large coastal power plants, including Mundra Ultra Mega Thermal Power Project and Mundra Thermal Power Station, were specifically designed around im-

[3]The copy of the revised presidential directive is provided in Annexure VI of the CIL Annual Report, 2014: https://d3u7ubx0okog7j.cloudfront.net/documents/2013-14˙8w1zW4u.pdf

[4]Define tapering/bridge linkage

[5]See Ministry of Coal's notification dtd. 26 July, 2013. https://coal.nic.in/sites/default/files/2020-01/260713_0.pdf

ported coal to circumvent India's constrained domestic supply system. These plants represent some of India's most efficient thermal capacity, employing supercritical technology and achieving superior environmental performance compared to older domestic coal-based units.

Import-based plants initially enjoyed significant advantages, with port locations eliminating India's substantial inland transportation costs that can effectively double fuel costs for plants located 1000 km from mine sources. During competitive bidding in the mid-to-late 2000s, these plants offered tariffs competitive with domestic coal-based plants, making them attractive to state electricity boards. However, their competitive position deteriorated sharply due to exogenous policy changes. In 2014, the Indonesian government mandated that coal exports be priced at international benchmark rates rather than previously negotiated long-term contract prices, substantially increasing fuel costs for Indian import-based plants (Taneja (2013)).

This policy shift created a supply-demand mismatch where plants were locked into fixed-price Power Purchase Agreements based on original competitive bids but now faced substantially higher input costs. While many plants secured temporary compensatory tariffs—Mundra UMTPP and Mundra TPS received increases of 0.52 and 0.41 INR/kWh respectively—the Supreme Court struck down this practice in 2017. Till this day, these plants produce electricity with imported coal, paying international benchmark prices. The combined effect of domestic coal access barriers and import price volatility has left some of India's most efficient thermal plants operating at reduced capacity factors, contributing to the systematic underutilization of cleaner generation capacity.

2017 SHAKTI Policy: The Scheme for Harnessing and Allocating Koyala Transparently in India introduced auction-based allocation for private generators while maintaining CIL's monopoly structure (Ministry of Coal (2017)). SHAKTI closed the LoA-FSA route, requiring private sector plants without linkages but with PPAs to bid through discount offers on electricity prices. In stark contrast, the policy explicitly maintained preferential treatment for public sector plants, which continued receiving administrative allocations at notified prices without participating in any competitive process. This dual treatment within the same policy framework perpetuated competitive disadvantages facing private generators.

Access to Coal Deposits and Captive Mining: Beyond CIL's direct coal supply, the evolution of captive mining policies created the most dramatic discrimination

between public and private generators. In 2007, amid shortfalls in domestic coal production and spikes in imported coal prices, industry appeals to the Prime Minister (who was also the Coal Minister) led to generous allocations of coal blocks on a first-come, first-serve basis at nominal prices, regardless of mining experience. These allocations later became controversial when the Comptroller and Auditor General identified an opportunity-based loss of Rs. 185,591.34 crore to the exchequer in what became known as "coalgate" (Comptroller and Auditor General of India (2012)).

The 2014 Supreme Court verdict cancelling 204 out of 214 allocated coal blocks created immediate financial distress for private power developers who had invested substantial capital based on securing low-cost captive coal. The cancellation eliminated low-cost coal supplies—increases from CIL's benchmark of 2,300 INR/tonne to 4,000 INR/tonne for market-sourced coal would raise variable costs by 1 INR/kWh, a 73 percent increase. Only 14 blocks were spared, including 12 allocated to NTPC and Sasan UMPP blocks.

The government's response through the Coal Mines (Special Provisions) Act, 2015, institutionalized the most explicit discrimination yet: mandating competitive auctions for private sector allocations while maintaining administrative "allotment" for public sector entities at nominal prices of INR 100/tonne for the power sector[6]. Private generators participated in competitive auctions with substantial premiums[7]—Monnet Power Company Ltd secured the Utkal C-block (with 123.86 million tonnes reserves) at INR 770 per tonne, excluding mining costs[8]. In contrast, public entities like NTPC received larger blocks at nominal prices, accessing blocks with geological reserves exceeding 7.3 billion tonnes. Most private blocks were sized under 10 MT, limiting economies of scale. By December 2018, only three private sector power plants were producing from their allotted captive mines(Central Electricity Authority (2018-2023b)), highlighting the practical difficulties private generators faced in developing these competitively acquired assets and the limited success of the revised auction system in providing viable alternatives for private sector coal access.[9].

[6]Economic Times (2015).https://economictimes.indiatimes.com/industry/indl-goods/svs/metals-mining/psus-to-pay-rs-100-150-per-ton-on-coal-besides-upfront-payment/articleshow/45979075.cms

[7]By August, 2015, 31 mines were auctioned off to the private sector, 9 of which were for power end-users

[8]Economic Times (2015).https://economictimes.indiatimes.com/industry/indl-goods/svs/metals-mining/monnet-power-company-bags-utkal-c-coal-block-for-rs-9537-22-crore/articleshow/46505049.cms

[9]These power plants include Sasan UMPP, Budge Budge TPS and Nigri TPP.

3.3.2 The Fundamental Distortion: Absence of Efficiently Functioning Coal Markets

These policy regimes - both in CIL supply and captive mining - have created systematic disadvantages for private generators through multiple mechanisms: timing of entry (post-2007 stricter regimes), explicit policy differentiation (public vs private treatment under SHAKTI and captive mining), and implicit advantages (composition of allocation committees, priority in shortages). Public sector plants receive administrative coal allocations with higher supply guarantees, better quality assurance, and priority during shortages. In contrast, private generators must participate in competitive auctions, face lower supply guarantees, and encounter more stringent performance requirements. This differential treatment forces private generators to rely on imported coal, significantly impacting their marginal costs and production levels.

3.4 Data

3.4.1 Electricity Generation and Outages

The Central Electricity Authority (CEA) publishes Daily Generation Reports, which include daily operational capacity and observed production for all utility-scale fossil, hydroelectric, nuclear and renewable power plants in India. I build an unbalanced panel combining the daily reports at the level of generator and day for the period of the study. The CEA's Daily Outage Reports provide the amount of capacity under outage for each plant on each day, except for non-hydro renewables. The reports also mention the reason for reported outage including fuel shortage.

3.4.2 Contract market data

I collect power plant capacity allocations to each state and the corresponding tariffs (both variable and fixed) for all power plants in the country from the Ministry of Power's MERIT portal. The portal reports the list of power plants contracted by the state utilities including the tariffs at which electricity was purchased from each. The dataset provides a complete description of the long term contract market including the tariffs (both fixed and variable costs) and contracted capacities.

The day-ahead market and short-term bilateral contracts comprise most of the short-term market. I use data on day-ahead market bids from the Indian Energy Exchange

(IEX). The IEX reports transmission region-wise demand and supply curves as well as market-clearing price and quantity at the exchange at 15-minute interval. The Indian electricity transmission network is composed of five transmission networks: North, Northeast, East, West, and South. Prices in the day-ahead market clear accounting for the transmission constraints between these five regions, resulting in region-specific prices. Only power plants with uncontracted capacity, and no fuel supply agreements (FSAs) with CIL participate in this market. I match thermal generators that participate in the market to the panel of generators above.

I also use short-term bilateral contract data collected from the Central Electricity Regulatory Commission (CERC). The contract data specifies the generating and buying entities, quantities traded as well as prices at the monthly level.

3.4.3 Wholesale electricity demand

I use daily data on the quantity of wholesale electricity purchased at the state level. The Power System Operation Corporation (POSOCO) operates the national grid and utilities report their electricity demand to the corporation daily. I utilise this data to construct a daily state-level panel of electricity demand.

3.4.4 Coal prices and quantities

The Central Electricity Authority's (CEA) Daily Coal Reports provide plant-day level data on coal receipts, stocks, and consumption for generation. I complement this with CEA's monthly coal reports that disaggregate receipts by source: Coal India Limited (CIL), Singareni Collieries Company Limited (SCCL), e-auctions, imports, and other sources. These reports also document monthly coal allocations specified in Fuel Supply Agreements (FSAs).

I construct source-specific coal prices using multiple data sources. For coal procured through FSAs (linkage-based coal), I use grade-specific notified prices from CIL and SCCL. Each plant is assigned a "minemouth" price based on the average grade (calorific value in kcal/kg) of coal it consumes. For e-auction coal, I utilize Ministry of Coal's monthly Cabinet reports that document scheme-specific premiums over notified prices and quantities. I restrict to two primary schemes through which power plants procure coal: Spot E-Auction and Special Forward E-Auction. For months with missing data, I supplement with average scheme-specific prices and quantities from the Ministry of Coal's Annual Reports.

For imported coal prices, I use monthly price series published by the Directorate General of Commercial Intelligence and Statistics (DGCIS). The DGCIS reports monthly quantity and value of steam coal imports (product code 27011920) by country of origin. I restrict to imports from Indonesia and South Africa as these two countries account for more than 90 percent of all imports. I validate them using a subsample of grade-specific coal import prices sourced from CoalMint.

4 Empirical evidence

This section provides both descriptive and reduced-form evidence that public sector generators receive higher coal allocation and receipts. pay lower coal prices compared to private generators and how that translates into a downward sloping marginal emissions curve. To provide empirical grounding to the structural model, I also provide evidence that private generators respond strongly to coal costs by reducing generation as the costs of coal procurement increase.

4.1 Efficient Private Generators Pay Higher Costs for Coal

Private generators show higher technical efficiency, measured in terms of heat rate (energy requirement for a power plant to produce one kWh of electricity). A lower heat rate means lower coal requirement per unit of output and hence, lower variable costs of electricity production. Appendix Figure A2 reveals two key patterns in heat rates. First, there is a clear vintage effect - plants commissioned before 2009, predominantly public sector, operate at heat rates exceeding 2500 kcal/kWh. Second, and more importantly, private generators consistently achieve better heat rates than public plants of the same vintage. Among plants commissioned during 2009-18, private plants operate at heat rates around 2200 kcal/kWh while comparable public plants require about 2400 kcal/kWh. This efficiency advantage persists across all post-2009 vintages, reflecting private plants' adoption of superior technology.

The technical efficiency advantage translates into substantial economic benefits under competitive coal pricing. Private generators consistently achieve heat rates around 2200 kcal/kWh compared to 2400-2500 kcal/kWh for public plants of comparable vintage. This 200–300 kcal/kWh efficiency advantage translates to INR 0.27–0.41 per kWh lower fuel costs at the import price of coal. More importantly for environmental policy, this efficiency difference directly translates to lower carbon emissions: public plants operating at 2600 kcal/kWh emit approximately 16% more CO_2 per kWh than

private plants at 2250 kcal/kWh. Such efficiency differences would normally provide private plants with substantial competitive advantages, enabling them to underbid public competitors in both long-term and daily dispatch decisions.

Figure 6 quantifies this discrimination across vintage cohorts of power plants. Pre-2009 plants, almost entirely public sector, received generous coal allocations exceeding 350 tonnes/MW. Although allocations declined for all plants commissioned after 2009, public generators maintained a significant advantage over private plants. This allocation pattern aligns with the policy evolution described above - the 2012 Presidential Directive and 2014 cancellation of coal blocks cemented these differences. Notably, substantial private entry (shown by capacity bars) occurred precisely during these post-2009 vintages when coal access became more restrictive and discriminatory.

Figure 6. Coal Allocation and Receipts



Figure shows coal allocations and receipts by plant ownership and vintage. Left panel plots CIL/SCCL coal allocation (tonnes/MW) against plant commissioning year for public (red) and private (blue) generators. Right panel shows actual coal receipts in the same format. Lines show average allocation/receipt by ownership while bars show installed capacity (GW, right axis) in each vintage bin. Vertical dashed lines mark key policy changes: New Coal Distribution Policy (2007), Presidential Directive on fuel supply agreements (2012), and SHAKTI policy (2017), explained in detail in section xx.

Figure 7. Coal Receipts by Source

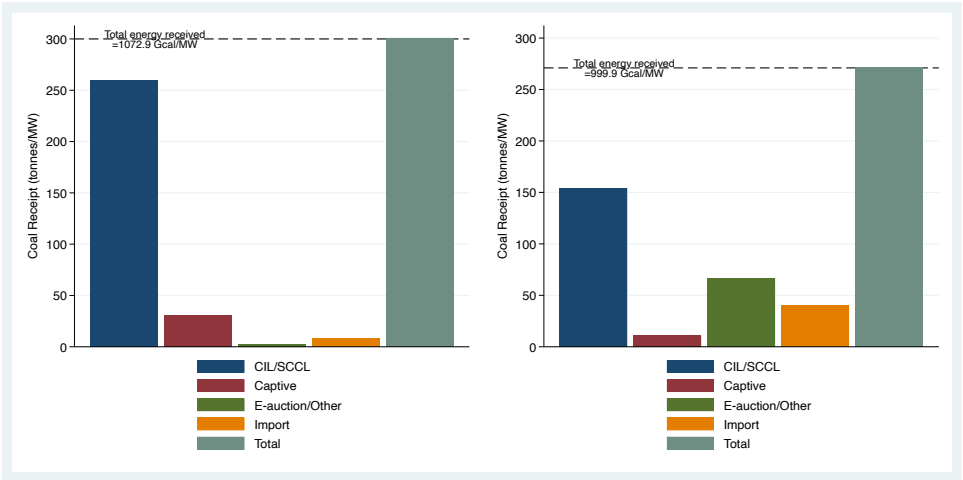


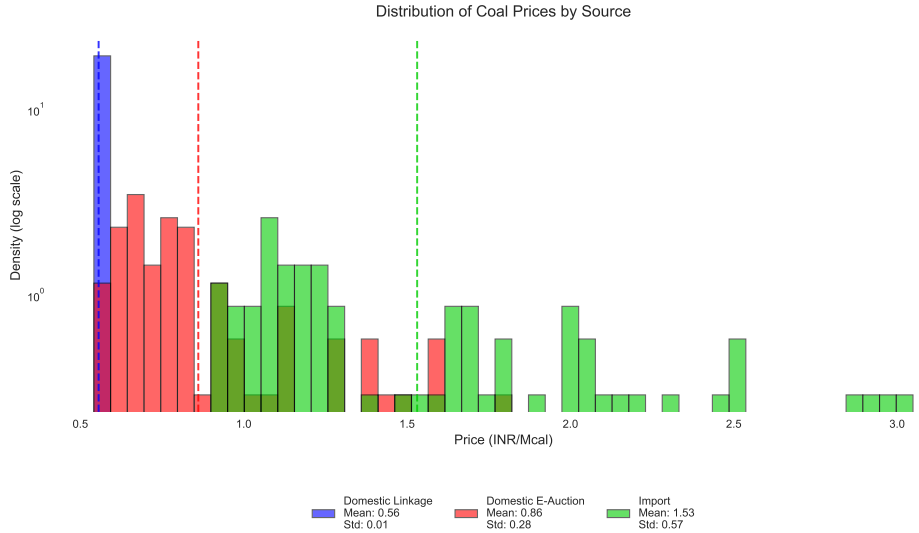
Figure shows the composition of coal receipts by source for public (left panel) and private (right panel) generators. Bars indicate average coal receipts (tonnes/MW) from different sources: CIL/SCCL linkage coal, captive mines, e-auction/other domestic sources, and imports. Total height represents aggregate coal receipts. Dashed lines show total energy received (Gcal/MW) after accounting for differences in coal quality across sources.

The right panel of Figure 6 demonstrates how these allocation differences translate into actual coal receipts. Public plants commissioned post-2009 receive around 250 tonnes/MW compared to about 170 tonnes/MW for private plants of the same vintage. Importantly, differences in receipts are far larger than differences in allocation, evidence that coal supply to public power plants is prioritised even when both public and private may have allocations with CIL. Figure 7 decomposes these receipts by source-public generators source most of their coal (over 250 tonnes/MW) directly from CIL/SCCL at subsidized prices, with minimal reliance on alternatives. In contrast, private generators receive significantly less CIL/SCCL coal (about 150 tonnes/MW) and must supplement with more expensive e-auction purchases, or imports. Even after accounting for the higher quality of imported coal, private plants receive significantly less energy per MW.

Figure 8 shows the cost implications of this supply discrimination. Expressing prices in energy-equivalent terms (INR/Mcal) to account for quality differences, private generators face substantially higher and more volatile input costs. Per unit of energy, e-auction coal costs about 55% more than coal bought at CIL's notified prices, and imported coal about 150% more. During 2018-2023, the average private generator paid about INR 3,400 per tonne of coal, against INR 2,200 for the average public generator, 58% more.

These price differentials create substantial cost disadvantages that overwhelm efficiency advantages. A thermal plant operating at a heat rate of 2200 kcal/kWh, burning coal of average quality, incurs fuel costs of INR 1.28 per kWh at the average price paid by public generators. At the average price paid by private generators, the same plant incurs INR 2.03 per kWh, 58% more. Such cost differentials systematically reallocate production from efficient private plants introduction large productive inefficiency.

Figure 8. Distribution of Coal Prices by Source



The figure shows the distribution of coal prices by source during 2018-2023. The figure plots histograms of prices in INR per Mcal for three coal sources: CIL/SCCL linkage coal, CIL e-auction coal, and imported coal. Prices are computed at the plant-month level and expressed in nominal INR. The y-axis uses a logarithmic scale to accommodate the large differences in density across price bins. Vertical dashed lines indicate the mean price for each source. Refer to section 5 for a detailed description of the data.

4.2 Differences in Costs of Coal Introduce Large Productive Inefficiency

Under efficient allocation, plants with lower marginal costs should operate at higher capacity factors, as they lie lower on the supply curve. For plants receiving fixed tariffs through long-term contracts, profit maximization implies generating up to the point where marginal cost equals the contracted tariff. For plants participating in day-ahead markets, absent other frictions, lower marginal costs translate to lower bid prices and higher dispatch probability. Both mechanisms predict that more efficient plants—those with lower heat rates and, under undistorted coal pricing, lower marginal costs—should achieve higher capacity utilization.

Coal market distortions invert this relationship by creating artificial cost disadvantages for efficient private plants. The resulting generation pattern then systematically favors less efficient plants, increasing aggregate generation costs and emissions for any given level of output.

Figure 9 provides direct evidence of this misallocation by comparing plant utilization under current and counterfactual coal pricing. The left panel shows how current coal market distortions create a wedge between coal costs for public and private generators: higher-cost bins contain disproportionately more private capacity, and these plants - despite superior heat rates - operate at systematically lower capacity factors. The pattern directly contradicts efficient allocation where the most efficient plants should achieve the highest utilization.

The right panel quantifies the extent of misallocation by showing how plants would be reordered along marginal cost bins if all generators faced uniform pricing at CIL's notified prices[10]. Under this counterfactual, 40% of the cheapest capacity would be privately owned, compared to only 18% under current pricing distortions. The differences in capacity utilization between panels (solid red lines) reveal the magnitude of misallocation: the most efficient generators currently operate at capacity factors around 0.4 when efficient allocation would yield utilization rates of approximately 0.55. This analysis shows that moving from the current distorted allocation to efficient dispatch would increase output from the most efficient plants while reducing generation from less efficient generators, lowering aggregate generation costs and emissions for any given level of electricity production.

[10]The choice of CIL's notified price as the common price is illustrative. The key insight about misallocation holds for any common price, whether it's the international market price or the notified price. Import-based plants appear less competitive under current pricing partly because they don't benefit from the implicit subsidy in domestic coal, itself a distortion that affects dispatch order.

Figure 9. Misallocation in electricity generation

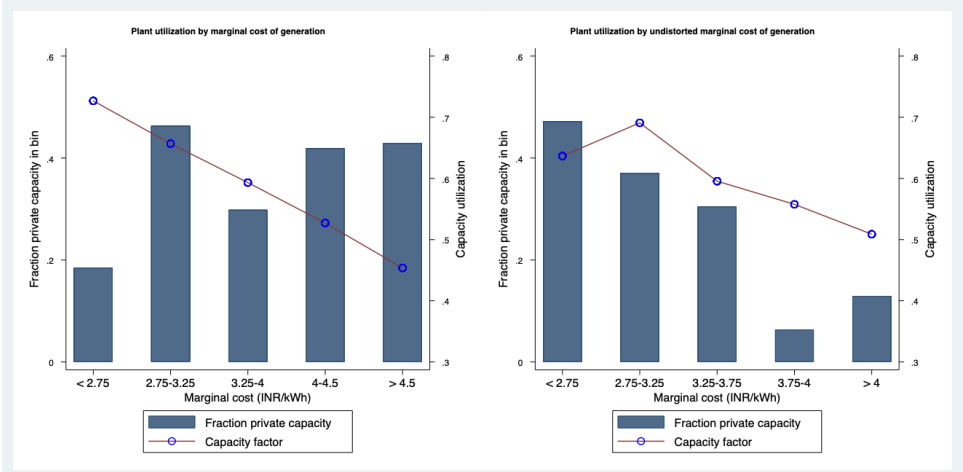


Figure shows the relationship between marginal cost of generation, ownership and plant utilization. Plants are grouped into cost bins (INR/kWh) on the x-axis. For each bin, bars show the fraction of total capacity in that bin that is privately owned (left y-axis) and the line shows average capacity utilization (right y-axis). Left panel uses observed marginal costs under current coal pricing. Right panel shows counterfactual marginal costs when all plants face CIL's grade-adjusted notified prices for coal.

4.3 Generator-level production responds to cost shocks

The observed misallocation from differences in capacity utilization between public and private plants could reflect several factors beyond coal market distortions. Private plants may face transmission constraints if they are located farther from load centers or have weaker grid connectivity, limiting their ability to supply power during peak demand periods. Technical factors such as different ramping capabilities, maintenance schedules, or plant availability could also contribute to utilization differences, particularly if private plant technology requires more frequent maintenance. Specifically to India, distribution companies (DISCOMs), which purchase most thermal power in India, face severe financial stress and payment delays that could influence procurement decisions—state-owned DISCOMs might preferentially purchase from public generators due to implicit government guarantees, political relationships, or simply because public plants are more willing to accept delayed payments given their own government backing. Additionally, differences in power contracting could matter: public plants typically have long-term contracts with assured off-take, while private plants may face greater exposure to short-term market volatility and demand uncertainty.

To establish that observed differences in capacity utilization reflect rational responses

to cost distortions rather than other factors, I exploit variation in coal cost shocks both across ownership types (extensive margin) and within private plants over time (intensive margin). The coal price shock provides plausibly exogenous variation in input costs, particularly salient for private generators given their reliance on market-priced imports and e-auction purchases.

I estimate three specifications that progressively illuminate the cost-generation relationship. My baseline specification is:

$$PLF_{ist} = \beta_1(lnCoalShock_t \times Private_i) + \beta_2 lnCoalShock_t + \gamma X_{st} + \alpha_i + \delta_t + \epsilon_{ist}$$

where PLF_{ist} is the capacity factor of plant i in state s and month-year t, $CoalShock_t$ is the monthly price of imported coal in INR/tonne, and $Private_i$ indicates private ownership. The coefficient of interest β_1 captures the differential response of private plants to coal price shocks. I include plant fixed effects α_i to absorb time-invariant plant characteristics and month and year fixed effects δ_t to control for seasonal patterns and aggregate trends. The vector X_{st} includes state-level electricity demand to control for local market conditions. All specifications are weighted by plant capacity to account for heterogeneous economic significance.

To better reflect the economic costs faced by plants, my second specification (column 2) transforms the coal price shock $CoalShock_t$ into generation costs in INR/kWh. To do this, I convert coal price in INR/tonne to INR/kWh using plant-specific heat rates and the average calorific value of coal, which varies by month. This specification provides coefficients that directly measure how generation responds to changes in unit costs of production.

Finally, I examine the relationship between actual marginal costs and generation for private plants (column 3). The specification instruments the marginal cost of generation (in INR / kWh) with imported coal price to address the potential endogeneity in marginal costs. I restrict this specification to private plants given the very limited variation in public plant costs, which would otherwise pose identification challenges.

The fixed effects systematically control for various confounding factors. Plant fixed effects α_i absorb all time-invariant characteristics including technological parameters (design heat rate, vintage, capacity), locational advantages (proximity to coal mines, ports, or demand centers), and systematic differences in operational efficiency. The inclusion of month fixed effects accounts for regular seasonal patterns in electricity

demand, coal quality, and plant maintenance schedules, while year fixed effects control for secular trends in the power sector including changes in merit order dispatch, improvements in grid integration, and evolution of coal markets.

The two-way fixed effects specification addresses concerns that plant-specific operational decisions might be correlated with unobserved determinants of efficiency or that aggregate shocks might differentially affect plants based on ownership. For instance, if private plants are systematically newer or located closer to ports, the plant fixed effects ensure that our estimates of β_1 are not conflating ownership effects with these correlated characteristics. Similarly, the time fixed effects ensure that our estimates are not driven by concurrent policy changes or market developments that might affect public and private plants differently.

Table 1 presents strong evidence that private generators systematically adjust generation in response to fuel cost shocks, while public plants show minimal response. In the baseline specification (Column 1), a 10% increase in coal prices leads to a 0.95 percentage point decrease in capacity factor for private plants relative to public plants, representing a 1.6% decline relative to the mean capacity factor of 0.58. Public plants show an economically and statistically insignificant 0.11 percentage point increase, consistent with their insulation from market price fluctuations.

The pattern persists when measuring cost shocks in generation-relevant units (Column 2), with nearly identical estimates confirming that the relationship reflects genuine cost sensitivity rather than spurious correlation with coal price movements. The instrumented marginal cost specification for private plants (Column 3) reveals even larger effects: a 10% increase in marginal costs reduces capacity factor by 3.13 percentage points, likely reflecting the full extent of cost exposure including plant-specific efficiency and coal quality factors.

The systematic responsiveness of private plants to cost shocks suggests that observed differences in capacity utilization reflect rational economic responses to distorted input prices rather than technical constraints or other factors. The lack of response among public plants confirms their insulation from market signals, consistent with their preferential access to subsidized coal.

Table 1. Cost shocks and generation

	(1) PLF	(2) PLF	(3) PLF
log Coal price shock (INR/tonne)	0.011		
	(0.013)		
Private (=1) x log Coal price shock (INR/tonne)	-0.095***		
	(0.011)		
log Coal price shock (INR/kWh)		0.015	
		(0.013)	
Private (=1) x log Coal price shock (INR/kWh)		-0.094***	
		(0.011)	
log Marginal cost (INR/kWh)			-0.313***
			(0.053)
Observations	11281.000	11281.000	4412.000
Dep var mean	0.580	0.580	0.573
Demand	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes
Month FEs	Yes	Yes	Yes
Station FEs	Yes	Yes	Yes

Table shows the relationship between coal price shocks and plant load factor (PLF). Columns (1) and (2) estimate the differential impact of coal price shocks on private plants using prices in INR/tonne and INR/kWh respectively. Column (3) estimates the direct relationship between instrumented marginal costs and PLF for private plants only. All specifications include plant, month, and year fixed effects, and control for state-level electricity demand. Regressions are weighted by plant capacity. Standard errors in parentheses are robust to heteroskedasticity.

4.4 Discussion: Implications for Carbon Pricing

The evidence in this section establishes the condition under which, in the framework of Section 2, a carbon tax is blunted: the coal allocation raises costs most for the cleanest generators. Private generators have lower heat rates, so they burn less coal and emit less CO_2 per unit of electricity; a heat rate of 2200 rather than 2500 kcal/kWh corresponds to about 12 percent lower emissions. Yet they pay about 58 percent more per tonne of coal than public generators. The generators that receive the least subsidized coal are therefore among the cleanest, and the distortion is negatively correlated with emissions intensity across generators.

This correlation weakens the mechanism through which a carbon tax reduces emissions. A tax raises each generator's cost in proportion to its emissions intensity, and reduces emissions by moving production from dirty generators to clean ones. When the distortion already lowers the costs of dirty generators relative to clean ones, the tax must first offset that cost advantage before it changes which generators are marginal and which are pushed out of production. The cost differences documented above are large relative to the carbon prices under discussion, so moderate taxes may change positions along the supply curve little. In the medium run, the same logic implies that a carbon tax may push clean generators that already operate at low capacity factors out of production, along with dirty ones.

The evidence also suggests an alternative to raising the carbon price. The distortion arises from how the coal subsidy is delivered at the margin, not from its size. A redesign that keeps generators' existing coal contracts, but gives every generator the same price of coal at the margin, would remove the distortion from the supply curve while leaving the subsidy's transfers in place. Production would move toward more efficient and cleaner generators without a carbon tax, and a carbon tax imposed afterwards would face no offsetting cost advantage.

How much the distortion reduces the effectiveness of a carbon tax, and how much the redesign would achieve, depends on the size of the distortion, its covariance with emissions intensity, and which generators are marginal on each day. The descriptive evidence measures the first two only partially, and cannot measure the third. The following sections measure the distortion directly at the generator level and build an empirical model of the electricity supply curve to quantify these effects.

5 Empirical Model and Estimation

5.1 Marginal Cost Construction

5.1.1 Coal-based generators

I estimate plant-specific marginal cost assuming that a plant's marginal cost does not vary with its level of output. To construct marginal costs of coal plants, I use the formula provided by the Central Electricity Regulatory Commission (CERC) Tariff Regulations 2019:

$$VC = (\frac{(SHR–SFC \times CVSF) \times LPPF}{CVPF} + SFC \times LPSFi)/(1–AUX)$$

where SHR is the station heat rate (kcal/kWh), SFC is the normative specific fuel oil consumption (ml/kWh), CVSF and CVPF are the calorific values of secondary and primary fuels, LPPF and LPSFi are the landed costs of primary and secondary fuels, and AUX is the auxiliary energy consumption.

The primary component driving variable cost is coal consumption, determined by station heat rate and coal quality. I estimate heat rates from plant-level daily coal consumption and generation data, supplemented with annual reports of coal calorific values, reported by CEA. While CERC regulations specify normative values for other parameters like auxiliary consumption and secondary fuel use, I rely on observed heat rates since they vary significantly across plants and vintages. Plants source coal through multiple channels: fuel supply agreements (FSAs) with Coal India Limited (CIL) or Singareni Collieries Company Limited (SCCL), e-auctions, captive mines, and imports. The delivered primary fuel cost LPPF is therefore a weighted average of source-specific coal prices, which I construct using:

- Grade-specific notified prices for FSA coal
- Auction premia over notified prices for e-auction coal (from Ministry of Coal reports)
- Import prices from DGCIS, adjusted for plant-specific reported coal quality
- Transportation costs based on rail distances from mines/ports to plants

I add government taxes, levies and surcharges to these costs to obtain the plant-specific "delivered price" of coal. For import-based plants, I assume zero inland

transportation costs as they are located close to the coast and receive their supply via ships.

I validate our constructed variable costs against the Merit Order Despatch of Electricity for Rejuvenation of Income and Transparency (MERIT) database[11] which reports costs based on power purchase agreements. For a consistent sample excluding lignite plants and those heavily reliant on imported coal, I find a correlation of 0.81 between my estimates and reported costs. More detailed methodology for construction of the marginal costs can be found in the Appendix.

5.1.2 Nuclear, gas and hydro based generators

Nuclear power plants in India have low variable costs, placing them near the beginning of the merit order. In the reported daily generation data, nuclear plants operate near capacity with an average plant load factor (PLF) of 0.96. Changes in the relative ordering of power plants along the supply curve in my counterfactuals do not affect nuclear utilization. Therefore, we place nuclear plants at the beginning of the supply curve to ensure they generate near capacity across all counterfactuals.

While hydro plants could theoretically serve as marginal generators during non-monsoon months (January through May and November through December), their actual dispatch reflects multiple constraints (Safiullah et al. (2017)). First, during monsoon months, generation is constrained by water availability. Second, hydro reservoirs serve multiple purposes including irrigation, drinking water supply, and flood control. Third, grid operators use non-merit order heuristics for hydro dispatch, as evidenced by the strong correlation between hydro generation and net demand (demand net of renewables and coal generation) (Sengupta et al. (2022a)). To account for these operational constraints, we place hydro generators at the bottom of the supply curve alongside renewables and nuclear but constrain their generation to match observed daily levels. This approach captures both water availability constraints and periods when hydro plants may serve as marginal generators.

The treatment of gas-based power plants requires careful consideration due to significant data limitations. Plant-specific prices for domestic gas and imported LNG are not publicly available at monthly frequency. Plant-level data on gas transportation and marketing costs are also unavailable. Monthly-level fuel consumption data

[11] Government of India, Ministry of Power, Merit Order Despatch of Electricity for Rejuvenation of Income and Transparency (MERIT) (2023); https://meritindia.in/

prevents reliable estimation of plant heat rates, particularly given the low capacity utilization of most plants. Moreover, a substantial share of gas capacity faces fuel shortages, as documented in CEA's Monthly Gas Reports, complicating the estimation of available capacity in counterfactuals.

Given these constraints, I assume gas generation remains at observed daily levels in my primary analysis. This approach is justified by three factors. First, gas represents a small share of both generation (2.4%) and capacity (5.0%). Second, gas-based generation faces significantly higher costs compared to coal. Third, my main counterfactuals examining coal market distortions are unlikely to significantly affect gas utilization given the large cost differences.

The cost disadvantage of gas relative to coal is substantial. During 2018-2023, domestic gas prices averaged 335.9 INR/MMBtu (range: 130-764 INR/MMBtu), while long-term imported LNG contracts averaged 750 INR/MMBtu (9-10 USD/MMBtu). At a conservative 50-50 allocation between domestic and imported gas, the variable cost of electricity for an average gas plant is approximately 7.0 INR/kWh[12]. This is more than twice the average variable cost of coal-based power plants in my sample (3.3 INR/kWh). In practice, the cost difference may be even larger as domestic gas supply is scarce and marginal requirements are typically met through spot LNG purchases. For instance, in April 2023, the lowest bid for gas-based peak summer supply was 13.7 INR/kWh, based on LNG pricing (Jain (2024)). At observed plant-level emissions factors, gas-based power plants become competitive with coal only at carbon taxes exceeding 100 USD/ton. Therefore, for carbon taxes between 0 and 100 USD/ton, we expect minimal fuel switching from coal to gas.

I do not explicitly model generation from renewable generators as these have zero marginal cost and are therefore dispatched to available capacity. I set renewable generation to observed values in all our counterfactuals. With these assumptions, the model explained in the next section simulates generation using all dispatchable generators (coal, gas, nuclear, hydro and renewables).

5.2 Measurement of the Coal Allocation Distortion

The marginal cost construction in Section 5.1 also delivers a direct measure of the coal allocation distortion at the generator-month level. For each plant i and month

[12]These calculations follow standard assumptions from https://www.fipi.org.in/assets/pdf/ep/Relative%20Costs%20of%20Power%20of%20Power%20Generation.pdf

t, I recompute marginal cost under a uniform-price benchmark in which every tonne of coal the generator burns is priced at the CIL (or SCCL) notified price per kcal, averaged across thermal grades and scaled to the plant's own coal quality. All other components of cost are unchanged: the plant's heat rate, auxiliary consumption and secondary fuel use, taxes and levies, and transport costs from the source mine or port. Denoting the benchmark cost by c^u_{it}, the distortion is the difference between the plant's observed marginal cost and this benchmark,

$$d_{it} = c_{it} - c^u_{it}, \qquad c_{it} = c^u_{it} + d_{it}. \tag{2}$$

Because the benchmark changes only the price of coal, d_{it} isolates the cost of rationed access to notified-price coal. Up to small differences in grade-level pricing, it equals the plant's heat rate times the price premium it pays on coal bought outside fuel supply agreement, weighted by the share of that coal in its receipts:

$$d_{it} \approx \frac{SHR_i}{1 - AUX_i}\Big[s^{eauc}_{it}\,(p^{eauc}_t - p^n_t) + s^{imp}_{it}\,(p^{imp}_t - p^n_t)\Big], \tag{3}$$

where s^{eauc}_{it} and s^{imp}_{it} are the shares of e-auction and imported coal in the plant's receipts, and p^{eauc}_t, p^{imp}_t and p^n_t are the e-auction, import and notified prices per kcal. Fuel prices, sourcing shares and freight are observed monthly, so the distortion varies at the generator-month level. Lignite plants, supplied from captive mines, have no distortion by construction.

This distortion measure is precisely the object the counterfactual removes. The no-distortion scenario in the counterfactuals simulates supply on c^u_{it}, so the change in production allocation between the status quo and the counterfactual is driven entirely by d_{it}. Plants with fuel supply agreements for most of their coal have distortions close to zero; while private generators that rely on e-auctions and imports face large distortion. Section 6 documents its distribution across generators and its relationship with emissions intensity.

5.3 Empirical Model of Electricity Generation

I build an empirical model of India's daily electricity dispatch to deliver three empirical requirements: (I) direct measurement of private costs and coal subsidies at the generator level, (II) an estimated relationship between generator costs and equilibrium production across the full set of generators, and (III) the ability to conduct

counterfactuals—carbon taxes or alternative subsidy designs—and re-solve for new equilibria. The model doesn't rely heavily on assumptions around generator incentives or conduct, rather, it uses detailed micro-data and regulation to empirically represent actual generator incentives and the dispatch mechanism used by electricity grid operators, estimated and validated against daily plant-level data for over 200 plants and 1,000 days. No free parameters are estimated to fit outcomes: costs and capacities are directly observed.

5.3.1 Economic Environment

Let i index generating plants, s index states, and t index days. For each plant-day, I observe marginal cost $c_{i,t}$ (constructed as described in Section 5.1), maximum available capacity $\bar{q}_{i,t}$ (net of maintenance outages and equipment failures), and contracted capacity $\bar{q}^{PPA}_{i,s,t}$ for plants with long-term contracts with state s.

Market structure. Two mechanisms govern how electricity is bought and sold. The first is long-term Power Purchase Agreements (PPAs): state electricity distribution utilities ("discoms") contract with individual plants, locking in dedicated capacity for 25-year terms. Each PPA specifies a variable charge—a fuel cost pass-through per kWh consumed—and a fixed charge covering capital costs. The variable charge is the operative constraint for short-run production decisions: a plant generates under its PPA only when its marginal cost falls below this contracted tariff. Since 2006, privately owned plants set variable charges through competitive bidding, with a share indexed semiannually to a CEA-published coal price series. The second mechanism is a short-term spot market (power exchanges and bilateral contracts) that clears the remaining 12% of demand at competitive prices. I extract plant-state allocation fractions and variable tariffs (both fixed and indexed components) for each plant-state pair from the MERIT India database and Central Electricity Authority capacity allocation records.

The critical institutional feature of PPAs is *exclusivity*: contracted capacity is locked to its designated buyer state. A plant serving State A cannot redirect surplus capacity to State B, even if State B faces unmet demand and State A has excess supply. This constraint is not a modeling choice, it is the legal structure of the PPA arrangement, enforced through Long Term Open Access rights that guarantee transmission capacity only for the specified source-destination pair. When a contracted state's demand falls short of contracted supply, the plant's surplus capacity sits idle.

Demand. I treat state-level demand as perfectly inelastic in the short run. This assumption is supported by two observations. First, end-users face regulated retail tariffs adjusted only annually, creating a wedge between wholesale market prices and the retail prices that drive consumption decisions. The price signal that would generate demand responsiveness simply does not reach consumers in the short run. Second, I estimate demand elasticity directly and find estimates statistically indistinguishable from zero. Under inelastic demand, the model's task is purely a supply-side allocation problem: assign a fixed quantity of generation across plants at minimum system cost, subject to institutional constraints.

Distortions enter through costs. Coal market distortions enter the model exclusively through generator-level marginal costs $c_{i,t}$. For plants with Fuel Supply Agreements (FSAs) with Coal India Limited, the fuel cost component of $c_{i,t}$ reflects subsidized domestic coal prices (approximately 40% of international benchmarks). For plants sourcing from spot auctions or imports, $c_{i,t}$ reflects market prices 1.5 to 2.5 times higher per unit of energy. Because the equilibrium generator level production depends directly on $c_{i,t}$, the subsidy structure shapes which plants generate at the margin. Counterfactual analysis then reduces to modifying $c_{i,t}$ and re-solving the same optimization: a carbon tax at rate τ enters as $c_{i,t} \to c_{i,t} + e_i \cdot \tau$, and coal price deregulation enters as $c_{i,t} \to c^*_{i,t}$, replacing subsidized fuel costs with market prices. Because both instruments operate through the same cost channel, the model delivers structurally consistent comparisons across policy experiments.

5.3.2 Two-Stage Model Structure

The dispatch mechanism has a two-stage economic structure that mirrors the actual institutional process.

Stage 1: Generator availability decision. Each generator with a PPA decides whether to declare capacity available to the grid operator by comparing its marginal cost against its contracted variable tariff $\bar{p}_i^{PPA}$: the plant reports available if $c_{i,t} \leq \bar{p}_i^{PPA}$ and declares unavailability otherwise. When marginal costs persistently exceed the indexed tariff—as occurs for private plants that source coal from expensive imports or spot auctions—plants may seek PPA renegotiation. This withholding decision is the generator's only short-run strategic lever; once capacity is declared available, the grid operator's optimization determines dispatch quantities. Merchant

plants (those without PPAs) always report capacity as available since they compete in the spot market at realized market-clearing prices rather than pre-committed tariffs.

Stage 2: System operator dispatch optimization. Given the set of available plants and capacity after Stage 1, the grid operator solves a cost-minimization linear program each day. The problem is:

$$\min_{\{q_{i,s,t} \geq 0\}} \sum_{i,s} c_{i,t} \cdot q_{i,s,t} \tag{4}$$

subject to:

$$\sum_{i} q_{i,s,t} = D_{s,t} \quad \forall s \qquad \text{(meet state demand)} \tag{5}$$

$$\sum_{s} q_{i,s,t} \leq \bar{q}_{i,t} \quad \forall i \qquad \text{(plant capacity)} \tag{6}$$

$$q_{i,s,t} \leq \bar{q}^{PPA}_{i,s,t} \quad \forall i, s \qquad \text{(PPA allocation)} \tag{7}$$

Constraint (5) requires that each state's demand is met exactly, reflecting inelastic demand and the grid operator's obligation to maintain supply-demand balance at the state level. Constraint (6) limits each plant to its physically available capacity. Constraint (7) is the critical institutional constraint: PPA plant i can supply at most $\bar{q}^{PPA}_{i,s,t}$ to state s—it cannot sell beyond its contracted allocation and, crucially, cannot serve states with which it has no contract ($\bar{q}^{PPA}_{i,s,t} = 0$ for non-contracted state pairs). Merchant plants face no such restriction ($\bar{q}^{PPA}_{i,s,t} = \bar{q}_{i,t}$ for all s), so the optimization freely allocates their capacity across any state with residual demand; these plants effectively compete in a national merit order to clear the aggregate residual demand pooled across states.

The solution to this linear program implements merit-order dispatch within the constraints of the contract structure: the cheapest available units are dispatched first, subject to state-level demand obligations and PPA exclusivity. The shadow price on constraint (5) is the market-clearing price: the marginal cost of the last (most expensive) unit dispatched to meet a given state's demand on a given day.

Key friction: stranded PPA capacity. Constraint (7) generates an empirically important source of allocative inefficiency. When a contracted plant has available ca-

pacity beyond what its dedicated state demands, that surplus cannot be redirected—it remains idle even if adjacent states face shortages and are drawing on more expensive plants. This is not a market failure in the sense of irrational behavior; it is an equilibrium consequence of the contracting arrangement, where long-term supply security motivates exclusivity terms. The practical implication is that first-best merit-order dispatch is not achieved even absent coal subsidies. Coal market distortions compound this inefficiency: by artificially reducing the private cost of high-emission public plants, they further entrench their position in the dispatch order, amplifying the welfare losses relative to the undistorted counterfactual (Jha et al. (2021)).

5.3.3 Model Validation

The model's inputs: available capacities $\bar{q}_{i,t}$ and marginal costs $c_{i,t}$, are directly observed or constructed from primary data, as described in Section 5.1. I validate the model against daily plant-level generation for over 200 plants and more than 1,000 days. The model explains 85% of the variation in actual output at the daily plant level ($R^2 = 0.85$), demonstrating that the institutional constraints captured by the optimization—PPA exclusivity, Stage 1 withholding decisions, and national merchant merit-order dispatch—are the primary determinants of observed generation patterns. The model correctly replicates relative utilization rates by ownership type (public versus private) and captures seasonal generation variation. As an additional validation of cost inputs, I compare constructed variable costs against costs that plants self-report to grid operators under their PPAs (from the MERIT database); for a consistent sample excluding lignite plants and those heavily dependent on imported coal, the correlation between constructed and reported variable costs is 0.81. Since the model contains no free parameters fitted to generation outcomes, this fit reflects the adequacy of the institutional representation, not sample overfitting. The remaining variation reflects hour-to-hour adjustments, strategic outage reporting, and non-cost dispatch that the daily model does not attempt to capture.

Limitations. The model abstracts from technical ramping constraints that prevent instantaneous output adjustment, hour-to-hour variation in transmission capacity (the model operates at daily frequency), and strategic behavior in availability reporting or PPA renegotiation. It also treats transmission as uncongested for non-PPA capacity, which is supported by Central Electricity Authority data showing transmission-related outages constitute less than 1% of total capacity outages during

the study period, and by evidence of no meaningful congestion for short-term bilateral and exchange-traded power during our study period (Athawale and Felder (2023)). These abstractions are standard for daily-frequency structural dispatch models (Fabrizio et al. (2007); Cicala (2022)). The primary aim is not high-temporal-resolution dispatch prediction but production reallocation across plants under alternative cost structures—the validation target relevant for counterfactual credibility—for which daily-level fit at $R^2 = 0.85$ is appropriate.

5.4 Counterfactual Analysis

Every counterfactual changes a single economic primitive, generators' marginal costs, and solves the model again for each day. Contracts, regulated prices, available capacity and demand are held at their observed values. The analysis is short run: it abstracts from generator entry and exit, investment in new capacity, and changes in electricity demand. Section 2 discusses how the same mechanism operates in the medium and long run. All counterfactuals are simulated for July 2021 to December 2023.

The first counterfactual is a redesign of the coal subsidy. Each generator supplies at its cost under the uniform price benchmark of Section 5.2, c^u_{it}, in which all of its coal is priced at the CIL or SCCL notified price per kcal for the coal quality it burns. This sets the distortion d_{it} to zero for every generator, while leaving its heat rate, transport costs and other cost components unchanged. Section 6.6 describes a compensatory tariff that implements this redesign: generators keep their existing fuel supply agreements, and the government compensates them for the difference between the market and notified price of any additional coal they buy, so that every generator faces the notified price per kcal at the margin.

The second set of counterfactuals imposes carbon taxes of \$10 to \$100 per tonne of CO_2, both on the current subsidy design and after the redesign. A tax at rate τ raises generator i's marginal cost to $c_{it} + \tau e_i$ under the current design, and to $c^u_{it} + \tau e_i$ after the redesign. Comparing the two schedules shows how much the current design reduces the effectiveness of a carbon tax. As an efficient benchmark, I also simulate the first best, which combines the redesign with a carbon tax of \$190 per tonne, the social cost of carbon.

To compute emissions, I multiply each plant's heat rate by the carbon content of its

coal. Plant-level gross calorific values come from CEA[13], while carbon content values use production-weighted averages from Srinivasan et al. (2018) for domestic coal and source-specific values for imported coal. Each counterfactual yields daily generation for every plant, and hence emissions, fuel use and transport costs, which Section 6 compares across policies.

6 Results

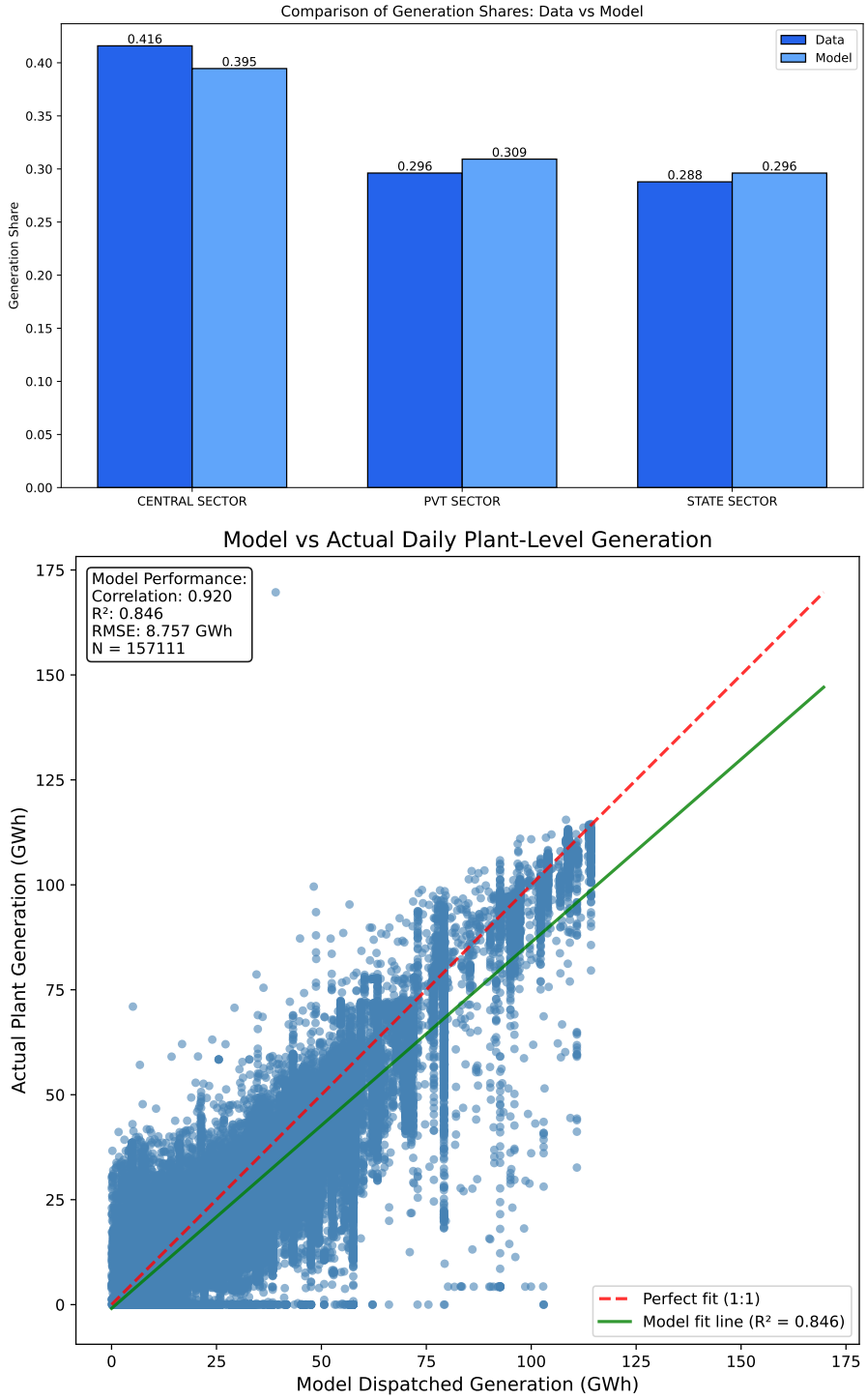
In this section, I present results focusing on the post-Covid period from July 2021 to December 2023. This is the period in which coal market distortions became particularly binding. Before this period, two factors effectively nullified the distortions: historically low electricity demand due to Covid-19 restrictions meant domestic coal was sufficient for most generation needs, while record low international coal prices provided a competitive alternative for plants requiring imports. Post-July 2021, the divergence between private and public plant costs widened substantially as demand recovered and international prices rose sharply, causing private plants' PLF to fall 5 percentage points below public plants despite a 13.5 percent increase in overall electricity demand. The demand patterns observed during this period better reflect both current market conditions and projected near-term trends in India's power sector (International Energy Agency (2023)). This makes the results more relevant for understanding ongoing and future inefficiencies in coal allocation.

6.1 Model fit

The model's predictions closely match the data along two dimensions: generation shares by ownership, and daily generation at each power plant. Figure 10 compares the two. The model explains 85 percent of the variation in daily production at the plant level ($R^2 = 0.85$), and the correlation between predicted and actual generation is 0.92. It also reproduces generation shares by ownership to within 2 percentage points: 39.5 percent predicted against 41.6 percent actual for the central sector, 29.6 against 28.8 percent for the state sector, and 30.9 against 29.6 percent for the private sector.

[13]Dataset compiled by Central Electricity Authority (CEA) containing CO_2 emissions data for all grid-connected power stations in India: https://cea.nic.in/cdm-co2-baseline-database/?lang=en

Figure 10. Comparison of Generation Shares: Data vs Model



Top panel shows aggregate generation shares by ownership type (central, state, private) comparing model predictions to actual data. Bottom panel shows scatter plot of daily plant-level generation (GWh) comparing model predictions (x-axis) to observed generation (y-axis), with the 45-degree line indicating perfect prediction.

The model also reproduces the contract structure of the market: 82.4 percent of simulated electricity flows occur within PPA obligations, against 88 percent in the data (Central Electricity Regulatory Commission (CERC)). Because contracted capacity comes with transmission rights, and short-term markets show no meaningful congestion, the simulated allocations respect the transmission constraints of the grid.

6.2 The Distortion across Generators

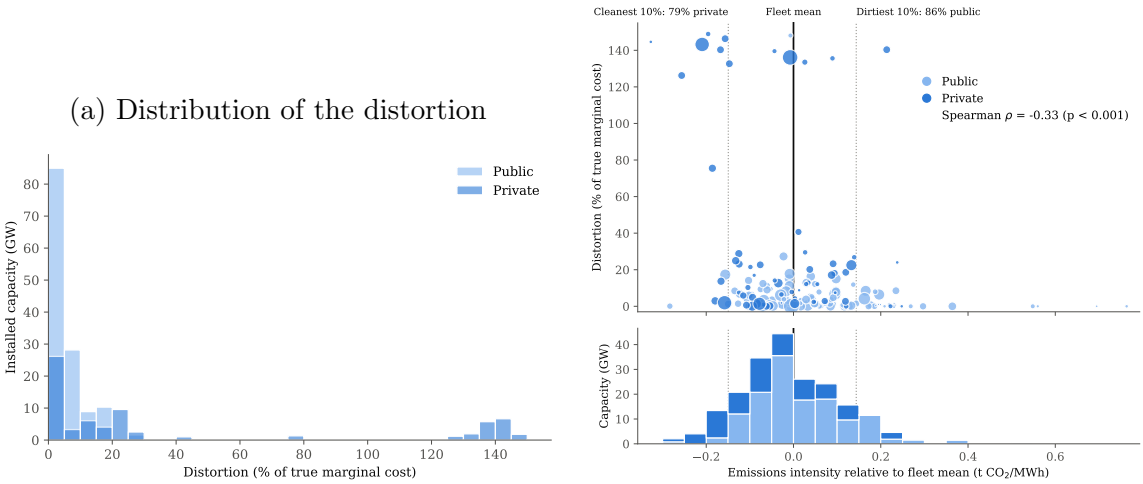
The theory in Section 2 predicts that a pre-existing distortion blunts carbon pricing when it is negatively correlated with emissions intensity: when the generators that the distortion penalizes are the cleanest ones. The direct measure of the distortion in Section 5.2 allows me to test this condition. Figure 11a shows its distribution across all coal generators. Public generators, which receive most of their coal under fuel supply agreements, face almost none: 63 percent of public capacity has a distortion below 5 percent of true marginal cost, and the capacity-weighted mean is 5.2 percent. Private generators face a distortion eight times larger on average (42.7 percent), and its distribution has a long right tail. A group of 13 generators, almost all private and supplied mainly with imported coal, face distortions above 100 percent of their true marginal cost.

Figure 11b plots each generator's distortion against its emissions intensity, measured relative to the capacity-weighted fleet mean of 1.013 tonnes of CO_2 per MWh. The two are negatively correlated (Spearman $\rho = -0.33$, $p < 0.001$). Most capacity lies close to the fleet mean, where the distortion varies little with emissions; the relationship is driven by the tails. The cleanest tenth of capacity is 79 percent private and faces an average distortion of INR 1.71 per kWh, nearly two-thirds of its true marginal cost. The dirtiest tenth is 86 percent public and faces INR 0.33 per kWh in distortion. The distortion therefore raises the private costs of the generators with the lowest social costs, which is the condition under which, in the theory, a carbon tax fails to reorder production to reflect these costs.

Figure 11. The coal allocation distortion across generators

(a) Distribution of the distortion

(b) Distortion and emissions intensity



The distortion is the difference between a generator's observed marginal cost and its marginal cost when all of its coal is priced at the CIL or SCCL notified price per kcal, holding coal quality, taxes and transport costs fixed (Section 5.2). It is expressed as a percentage of this undistorted (true) marginal cost and averaged over July 2021–December 2023. The sample covers 172 coal and lignite generators. Panel (a): installed capacity by distortion, in bins of 5 percentage points, shown separately for public (central and state) and private generators. Panel (b), top: each marker is a generator, with marker area proportional to installed capacity; emissions intensity is measured relative to the capacity-weighted fleet mean of 1.013 tonnes of CO_2 per MWh. The solid vertical line marks the fleet mean, and the dotted lines the capacity-weighted 10th and 90th percentiles of emissions intensity; the labels report the private and public shares of capacity in the cleanest and dirtiest tenths. The Spearman rank correlation between the distortion and emissions intensity is computed across generators. Panel (b), bottom: installed capacity by emissions intensity, in bins of 0.05 tonnes of CO_2 per MWh, stacked by ownership.

6.3 Marginal emissions curve under different policies

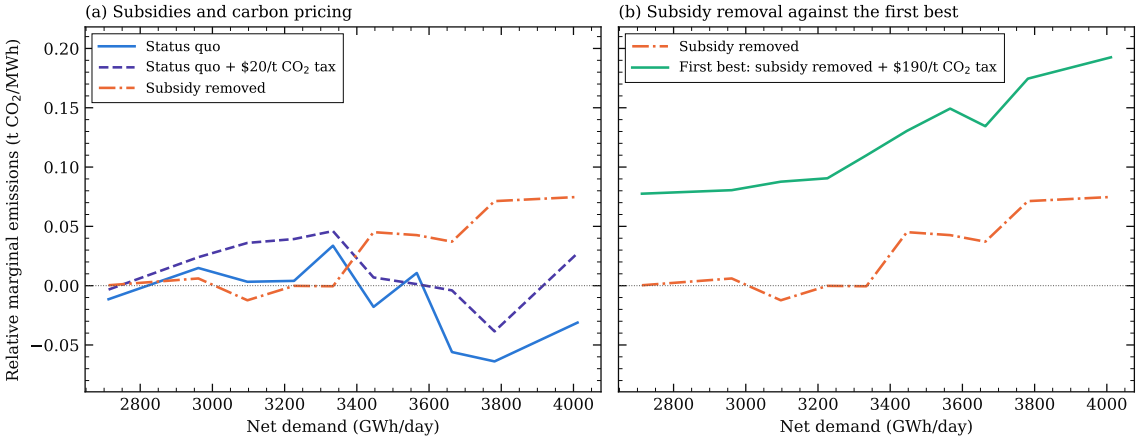
I compare four policies, each simulated with the empirical model over July 2021 to December 2023. The *status quo* allocates production on the generator's observed marginal costs. A carbon tax adds the tax times the generator's emissions intensity to its marginal cost; I report a tax of \$20 per tonne of CO_2 here and a range of taxes in Section 6.5. Removing the distortion re-orders generators at costs at the uniform notified price, c^u_{it}, as under the compensatory tariff described in Section 6.6. The *first best* removes the distortion and adds a \$190 tax, so that production allocates to reflect the true social costs.

For each policy, I recover the marginal emissions curve: the emissions intensity of the *average* generator that serves the last unit of demand, as a function of demand. On

each day, the marginal generator is the highest-cost generator that is dispatched, and I average its emissions intensity, relative to the fleet mean, within deciles of daily net demand.

The first-best curve is the efficient benchmark, recovered from the same model. It slopes upwards, and marginal emissions are high at every level of demand: the cleanest generators are always infra-marginal, so the generators that serve at the margin are progressively dirtier (Figure 12, panel b). The status quo curve has the opposite shape. Its marginal generator is close to the fleet mean at low demand levels and 5 percent cleaner than the fleet mean at high demand levels: clean generators operate at the margin while dirtier ones are infra-marginal. A $20 tax flattens the curve but still does not reverse the shape. Removing the distortion on the other hand takes the supply curve significantly closer to the first best: the curve slopes upward, and the marginal generators are on an average, dirtier, yet aggregate emissions fall, because the dirtiest generators now produce only when demand is high rather than in the base of the supply curve. The next subsection shows how generators move along the supply curve to produce these shapes. Removing the distortion also produces an upward-sloping curve in specific markets, for instance, in five of the six states with the largest demand and private capacity (Appendix Figure C1).

Figure 12. Marginal emissions curves under different policies



Each curve plots the emissions intensity of the marginal generator against daily net demand (demand net of renewable, hydro and gas generation), simulated with the model. On each day, the marginal generator is the highest-cost generator serving the demand nationally, with costs including any carbon tax; ties are weighted by generation. Emissions intensity is measured relative to the capacity-weighted fleet mean of coal and lignite generators (1.013 tonnes of CO_2 per MWh) and averaged within deciles of daily net demand; each point is plotted at the mean demand of its decile. Panel (a) compares the status quo, a $20 per tonne carbon tax on the status quo, and removal of the distortion. Panel (b) compares removal of the distortion with the first best, which removes the distortion and adds a carbon tax of $190 per tonne, the social cost of carbon. July 2021–December 2023.

6.4 Why the curves differ: who gets pushed out and who turns infra-marginal

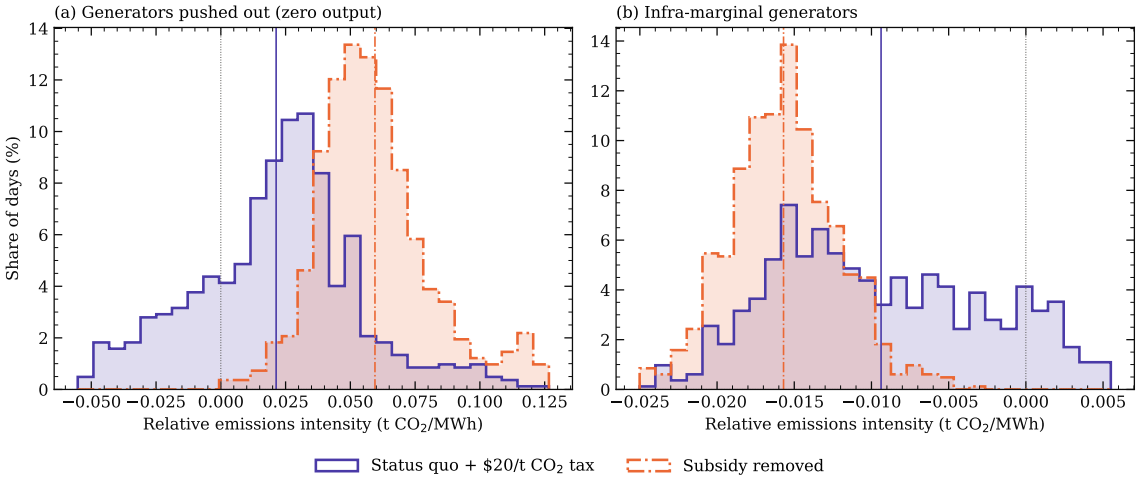
I use the $20 tax to illustrate the mechanism. Figure 13 compares, day by day, the emissions intensity of two groups of generators: generators that are available but produce zero, and infra-marginal generators that operate near-capacity. Under a $20 tax on the status quo, the capacity pushed out of production is on average only 2.1 percent dirtier than the fleet mean, and on 24 percent of the days it is cleaner. Removing the distortion pushes out capacity that is 5.9 percent dirtier than the fleet mean, and never cleaner, while the generation that remains in production is 1.6 percent cleaner than the fleet mean, against 0.9 percent under the tax. Removing the distortion therefore pushes the dirtiest generators out and pulls cleaner ones in.

Figure 14 shows the resulting reallocation across the emissions distribution. Under a $20 tax, generation shifts from dirtier to cleaner generators, but by small amounts: generators at least 15 percent cleaner than the fleet mean gain 1.3 GWh a day on average, and those at least 25 percent dirtier lose 3.2 GWh. Removing the distortion

moves far more production at both ends: the cleanest generators gain 11.8 GWh a day, almost all of it private, and the dirtiest lose 4.9 GWh, all of it public. Across the fleet, private generation rises by 24 percent (90 TWh a year) while state-owned generation falls by 21 percent (72 TWh) and central-government generation by 4 percent (18 TWh). Under the $20 tax, the corresponding changes are 2.4, −1.8 and −0.6 percent.

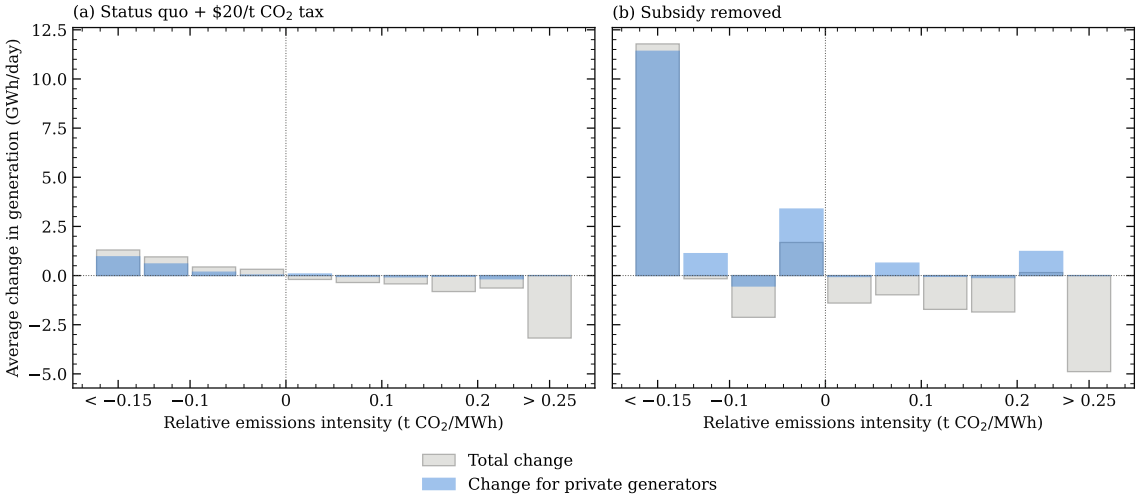
The emissions reduction comes from the dirty end of this reallocation. Removing the distortion lowers annual CO_2 emissions by 14.6 million tonnes, 1.2 percent of the sector's emissions. Lignite generation, the most emissions-intensive in the fleet, falls by 12 TWh a year, a reduction of 16.1 million tonnes. Coal generation rises by about as much to replace it, but because it shifts toward more efficient coal plants, coal emissions rise by only 1.5 million tonnes.

Figure 13. Emissions intensity of generators pushed out of production and of infra-marginal generators



Distributions across days of the national emissions intensity of two groups of coal and lignite generators, under a $20 per tonne carbon tax on the status quo and under removal of the distortion. Panel (a): generators with available capacity that produce zero MWh on that day, weighted by available capacity. Panel (b): infra-marginal generation, meaning all dispatched generation except that of each state's marginal generators (the highest-cost generators supplying the state on that day), weighted by generation. Emissions intensity is measured relative to the capacity-weighted fleet mean (1.013 tonnes of CO_2 per MWh). Vertical lines mark the mean of each distribution. July 2021–December 2023.

Figure 14. Reallocation of generation across the emissions-intensity distribution

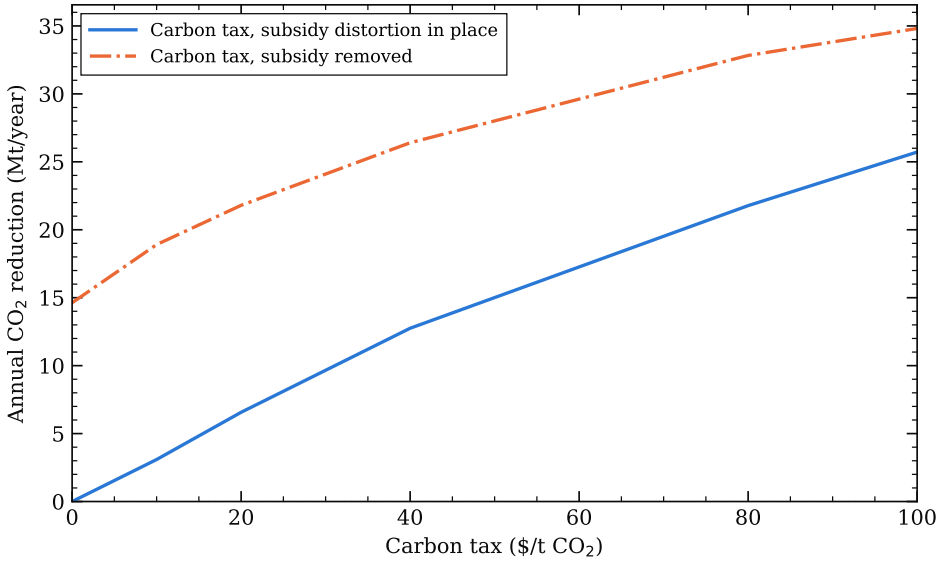


Each bar is the average change in daily generation per generator, relative to the status quo, across all generator-days within a bin of emissions intensity. Panel (a): a $20 per tonne carbon tax on the status quo. Panel (b): removal of the distortion. Emissions intensity is measured relative to the capacity-weighted fleet mean (1.013 tonnes of CO_2 per MWh), in bins of 0.05 tonnes; generators more than 0.15 tonnes cleaner or 0.25 tonnes dirtier than the fleet mean are pooled into the end bins. The gray bar is the total change; the blue bar is the contribution of private generators, so the remainder is the contribution of public (central and state) generators. Coal and lignite generators with available capacity, July 2021–December 2023.

6.5 Carbon taxes in a distorted market

Figure 15 reports annual CO_2 reductions across carbon taxes from $0 to $100 per tonne, with and without the distortion. The estimated numbers are consistent with those reported in literature for Indian electricity sector (Sengupta et al. (2022b)). Removing the distortion alone reduces emissions by 14.6 million tonnes a year, 1.2 percent of the sector's emissions and more than a $40 tax achieves on the distorted market (12.8 million tonnes). Matching the reductions of a $40 tax combined with removing the distortion (26.4 million tonnes) requires a tax of $100 on the distorted market (25.7 million tonnes). At $40 per tonne, the distortion therefore costs 13.6 million tonnes of additional abatement forgone, worth $2.6 billion a year at the social cost of carbon. The foregone abatement shrinks at high tax rates because a large enough tax eventually overrides the cost advantage the distortion gives dirtier generators, as Section 6.6 shows directly: a $20 tax offsets about half of the cost gap the distortion creates between the cleanest and dirtiest generators, and a $40 tax roughly cancels it. Even a $100 tax on the distorted market reduces emissions by only 2.2 percent (2.9 percent with the distortion removed).

Figure 15. Emissions reductions from carbon taxes with and without the coal allocation distortion



Annual CO_2 emissions reductions (million tonnes per year) relative to the status quo without a carbon tax, for carbon taxes of \$0, 10, 20, 40, 80 and 100 per tonne. The solid line imposes the tax on the distorted market. The dash-dotted line imposes the tax after removing the distortion, with generators supplying at costs at the uniform notified price of coal (Section 5.2); at a zero tax, it shows the reduction from removing the distortion alone. Emissions are computed from simulated plant-level generation and emissions intensities (Section 5.4). Annual averages over July 2021–December 2023; status quo emissions are 1,195 million tonnes a year.

6.6 Welfare analysis

I evaluate the removal of the coal allocation distortion through a compensatory tariff that leaves the institutional structure of the coal market intact. Coal India's fuel supply agreements and notified prices are unchanged: each plant keeps its status-quo tonnage of linkage coal at the notified price. Any coal a plant needs beyond that tonnage is bought on the market, through e-auctions or imports, at a common price equal to the import-parity price adjusted for calorific value, and the government reimburses the gap between this market price and the notified price. India has used a similar mechanism. In May 2022, facing a shortage of domestic coal, the Ministry of Power invoked Section 11 of the Electricity Act to direct all plants designed to run on imported coal to operate at full capacity. Because their long-term contracts did not allow high import prices to be passed through, a committee of the Ministry of Power, the Central Electricity Authority and the Central Electricity Regulatory Commission set benchmark rates that covered the plants' fuel costs, and the direction

was renewed for the summer of 2023.[14] [15] Every generator therefore bids as if it paid the notified coal price per kcal, which removes the distortion from the supply curve without changing how coal is priced. The reform raises efficiency along private costs through two channels. First, production moves to plants that need less fuel per unit of electricity: annual coal and lignite consumption falls by 5.6 million tonnes of standard coal (5,000 kcal/kg), as efficient coal plants displace both less efficient coal plants and lignite. Second, production moves to plants that are cheaper to supply, coastal plants that receive imported coal by sea and inland plants closer to mines, which reduces transportation costs, specifically, rail freight. Valued at the international market price of coal per kcal, the relevant opportunity cost because India imports coal at the margin, these fuel and transport savings quantify the misallocation of production along private production costs. [16] The explicit transfer mechanism in this design impacts relevant stakeholders differently. Coal India resells excess linkage coal that inefficient plants no longer demand at the market price, a windfall for the government that owns it; the government compensates generators for all coal bought on the market; and consumers pay less for electricity because production costs reduce, and market price falls.

Figure 16 reports the annual welfare effects of each counterfactual policy relative to the status quo.[17] Removing the distortion saves $1.9 billion a year in fuel and transport costs: $0.7 billion from lower fuel consumption and $1.3 billion from lower freight costs. These efficiency gains are large, but modest relative to the transfers the subsidy reform sets in motion. The average market price of electricity falls from INR 5.3 to INR 4.1 per kWh, and consumers save $6.8 billion a year.[18] The government pays $10.4 billion in compensation to generators, of which Coal India's windfall from reselling 58 million tonnes of standard coal recovers $5.2 billion, for a net fiscal cost of

[14] Ministry of Power directions under Section 11 of the Electricity Act, 2003, dated 5 May 2022 and renewed for 16 March to 15 June 2023; see also CERC Order in Petition No. 128/MP/2022, https://cercind.gov.in/2022/orders/128-MP-2022.pdf.

[15] Linkage coal that a plant no longer procures because it has been pushed to be marginal by the policy, is resold by Coal India Limited at the market price. Lignite plants are supplied from captive mines and receive no compensation.

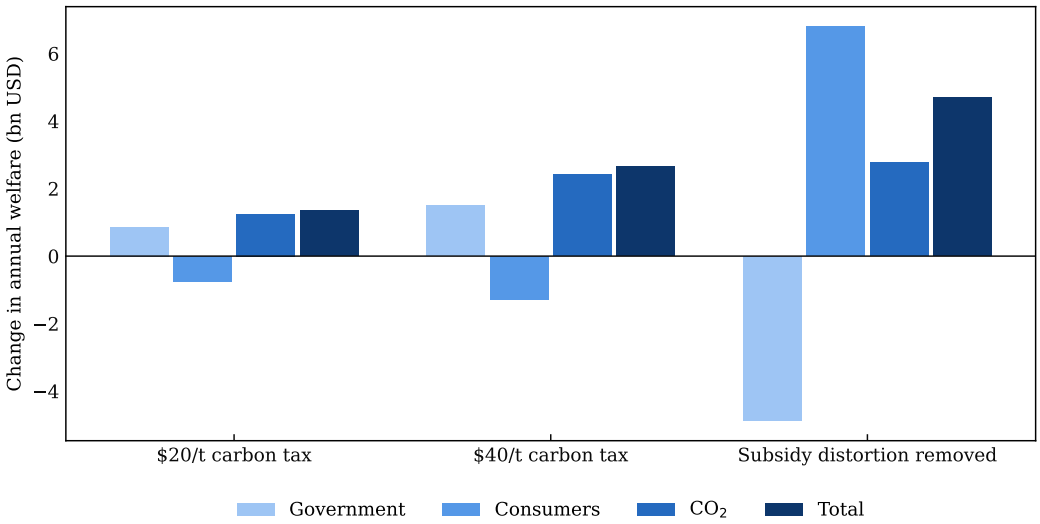
[16] I assume that coal and lignite have the same resource value per kcal and that Coal India's output is fixed, so the change in demand for linkage coal equals the change in imports. Freight per tonne is plant-specific and identical across scenarios; freight savings arise only because generation moves to plants with lower freight per MWh. Under these assumptions, the sum of changes in consumer, generator and government surplus equals the fuel and transport savings exactly.

[17] All values are annual averages over July 2021 to December 2023.

[18] Consumer gains are net of the change in generators' margins on non-contracted sales (−$0.15 billion). Contracted generation is paid its variable cost and earns no margin in any scenario.

$4.9 billion. The difference between consumers' gains and the government's net cost is exactly the efficiency gain. Removing the distortion also reduces CO_2 emissions by 14.6 million tonnes a year, worth $2.8 billion at a social cost of carbon of $190 per tonne, for a total annual welfare gain of $4.7 billion. Carbon taxes imposed on the distorted market achieve significantly less. A $20 per tonne tax raises annual welfare by $1.4 billion and a $40 tax by $2.6 billion, almost entirely through lower CO_2 emissions (6.6 and 12.8 million tonnes a year). Their efficiency gains are small, $0.1 and $0.2 billion, because a uniform tax on emissions doesn't meaningfully change the relative position of generators along the supply curve. Across coal and lignite generators, the dirtiest tenth of capacity emits about 50 percent more CO_2 per MWh than the cleanest tenth (1.24 against 0.83 tonnes), so a $20 tax widens their cost gap by INR 0.66 per kWh. The distortion works in the opposite direction and is larger: it raises the marginal cost of the cleanest tenth by INR 1.71 per kWh, against INR 0.33 per kWh for the dirtiest tenth, a gap of INR 1.38 per kWh. A $20 tax offsets only half of this gap, and even a $40 tax, at INR 1.33 per kWh, does little more than cancel it, leaving the cleanest generators no cheaper than the dirtiest. The distribution of gains also differs. Even with the entire tax revenue ($24 and $47 billion) recycled back, consumers lose $0.8 and $1.3 billion, because generation shifts toward plants that buy coal at market prices and pass these costs through. The same shift lets Coal India resell linkage coal at market prices, so the consumer loss reappears as a government gain of $0.9 and $1.5 billion. Removing the distortion thus yields 1.8 times the welfare gain of a $40 carbon tax and larger emissions reductions, by correcting the allocation of production rather than by raising the price of electricity.

Figure 16. Welfare effects of carbon taxes and of removing the coal subsidy distortion



Change in annual welfare (billion USD) relative to the status quo under a $20 and a $40 per tonne carbon tax on the distorted coal market, and under removal of the coal subsidy distortion through a compensatory tariff without a carbon tax. Government: compensation paid to generators, Coal India's windfall on resold linkage coal, changes in cess and import duty; carbon tax revenue is rebated in full to consumers and nets to zero. Consumers: change in electricity payments, net of the carbon tax rebate and of the change in generators' margins. CO_2: change in emissions valued at $190 per tonne. Total is the sum of the three components. Annual averages over July 2021–December 2023, computed using the conversion factor: INR 80 per USD.

7 Conclusion

This paper evaluated the impacts of the design of India's coal subsidies for the electricity sector on the effectiveness and welfare gains of carbon pricing. The current subsidy gives the cleanest and most efficient generators the highest price of coal at the margin, creating a negative correlation between the distortion and emissions intensity across generators. The distortion therefore acts like a negative carbon price: it shields the dirtiest generators from the carbon price signal, pushes clean generators to the margin and raises aggregate emissions. I measure the distortion for each generator with data and estimate an empirical model of daily electricity supply that captures regulated electricity prices, contracts and availability decisions of Indian generators to evaluate counterfactual policies.

Redesigning the subsidy, so that every generator faces the same price of coal at the margin while keeping its existing coal and electricity supply contracts, reduces CO_2 emissions by 14.6 million tonnes a year, more than a $40 per tonne carbon tax would

achieve under the current design. Because production moves to generators that burn less coal and are cheaper to supply, the redesign saves further $1.9 billion a year in fuel and transport costs, and its total welfare gain of $4.7 billion a year is 1.8 times that of the tax. The redesign also makes carbon pricing more effective. With the subsidy redesigned, a $20 tax achieves the reduction in emissions that requires an $80 tax under the status quo, and a $40 tax achieves the reduction that requires a $100 tax.

The results carry important lessons for countries that introduce carbon pricing on top of existing energy subsidies, as India proposes to do with its Carbon Credit Trading Scheme. Where subsidies are hard to remove, their design at the margin matters for climate policy in two ways. First, it determines the effectiveness of a carbon price, and with it the choice of instrument: a carbon tax is theoretically efficient, but a subsidy that shields dirty producers can absorb so much of its signal that a tax of a feasible size may not reach the emissions target at all. Second, it determines the cost of reaching a given target, because a carbon price that is less effective must be set higher to achieve the same reduction. The effect of a subsidy on a carbon price depends on a single statistic, the covariance between the distortion and emissions intensity across producers, which can be measured directly. A subsidy design that gives every producer the same price at the margin aligns production with true costs, and because it relies on administered prices and supply contracts that already exist, it requires no new regulatory or enforcement capacity. Correcting the design of existing subsidies is therefore a practical complement to carbon pricing in the economies where carbon pricing is now being introduced.

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A Appendix Tables and Figures

Appendix Figure A1. Electricty generation by source in major countries

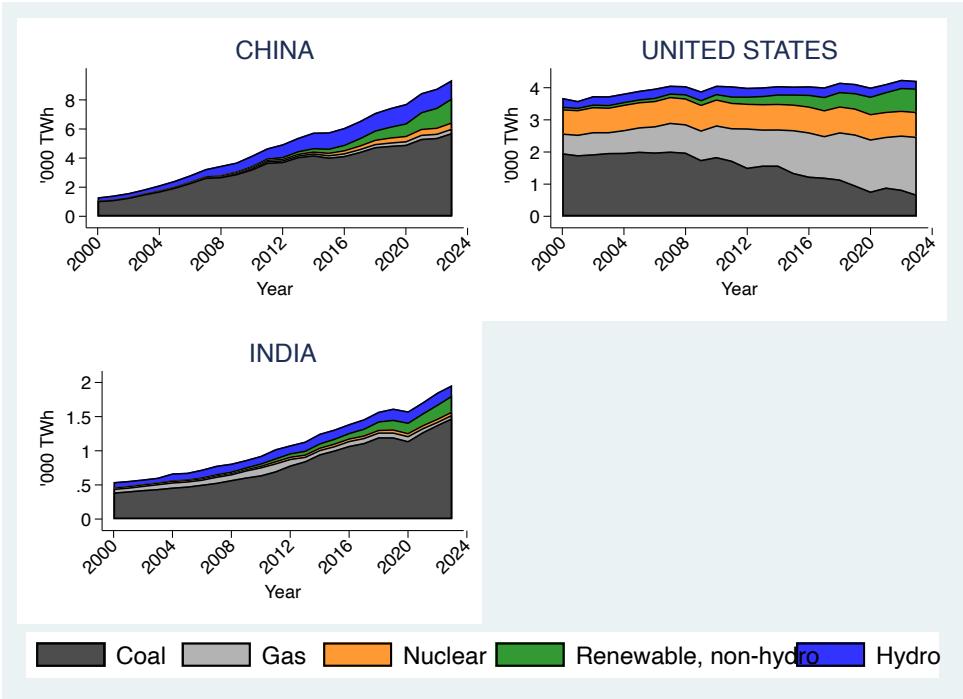


Figure shows electricity generation by source in China, United States, and India from 2000-2024. Generation is measured in thousand terawatt-hours (TWh) and broken down into five categories: coal (dark gray), gas (light gray), nuclear (orange), non-hydro renewables (green), and hydro (blue). Source: Ember (2024); Energy Institute - Statistical Review of World Energy (2024)

Appendix Figure A2. Plant Heat Rate by Ownership and Commissioning Year

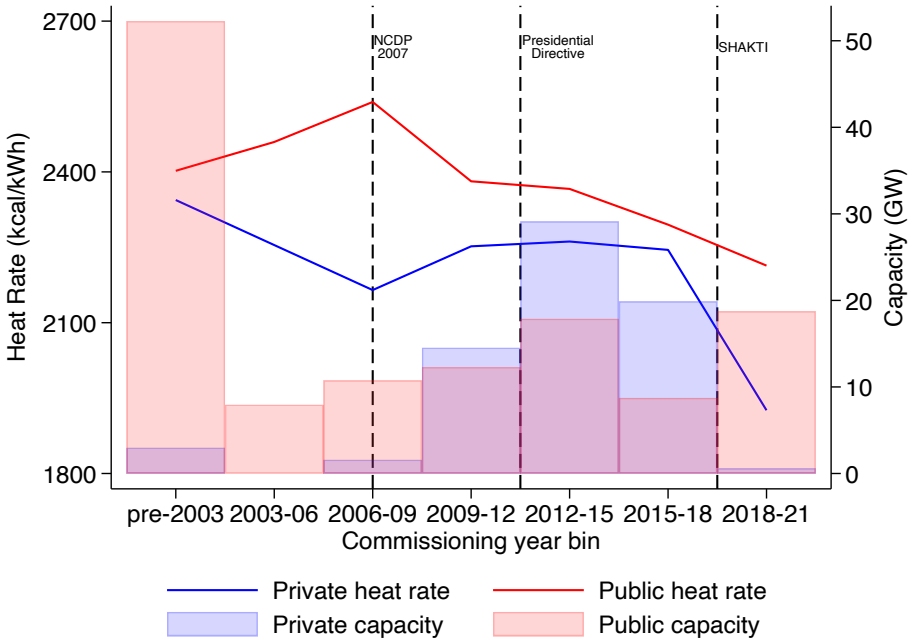


Figure shows plant heat rates (in kcal/kWh) by ownership and vintage. The graph plots heat rate against plant commissioning year for public (red) and private (blue) generators. Lines show average heat rate by ownership while bars show installed capacity (GW, right axis) in each vintage bin. Vertical dashed lines mark key policy changes: New Coal Distribution Policy (2007), Presidential Directive on fuel supply agreements (2012), and SHAKTI policy (2017), explained in detail in section 2.

B Appendix One: Variable cost estimation

The methodology for estimating plant-specific variable cost of electricity generation (VC), expressed in INR per kWh, is based on a formula provided by the Central Electricity Regulatory Commission (CERC) Tariff Regulations 2019. The formula is as follows:

$$VC = (\frac{(SHR–SFC \times CVSF) \times LPPF}{CVPF} + SFC \times LPSFi)/(1–AUX)$$

where:

SHR = Gross station heat rate, in kCal per kWh

SFC = Normative specific fuel oil consumption, in ml per kWh

$CVSF$ = Calorific value of secondary fuel, in kCal per ml

$LPPF$ = landed fuel cost of primary fuel, in INR per kg or cubic metre, as applicable

$CVPF$ = Gross calorific value of primary fuel, in kCal per kg or cubic metre, as applicable

$LPSFi$ = Landed cost of secondary fuel in INR per ml

AUX = Normative auxiliary energy consumption

The formula incorporates various factors that contribute to the variable cost of electricity generation, including the efficiency of the power plant (represented by the station heat rate), the costs and calorific values of both primary and secondary fuels, and the energy consumed by the plant's auxiliary systems. The CERC Tariff Regulations also specify normative assumptions to be used in determining the variable cost. These assumptions vary based on unit capacity, vintage, and type of primary fuel used in the power plant. I use these normative assumptions in the above formula, except for heat rate.[19] [20] I estimate heat rate from data on daily plant-level electricity generation and coal consumption and annually reported average gross calorific value of primary fuel.

[19] The regulations can be found here: https://www.cercind.gov.in/2019/regulation/Tariff%20Regulations-2019.pdf

[20] Assumed LPSFi = Rs. 50000/kL. This assumption is based on the range of values approved by State Electricity Regulatory Commissions (SERCs) for the period under consideration. These commissions approve the variable costs of power plants annually upon submission of tariff petitions, with approved LPSFi values typically ranging from 45,000 to 60,000 INR per kiloliter. For instance, see https://cserc.gov.in/upload/order/0-9-2023_6878-1692785214-.pdf and https://www.cercind.gov.in/2024/orders/445-GT-2020.pdf

The contribution of secondary fuel to the overall variable cost is marginal. The majority of the variable cost is derived from the consumption of the primary fuel, which in this case is coal. I explain the methodology for constructing the landed cost of coal, a crucial component of this calculation, in a separate section.

Comparison with MERIT database

Calculated variable costs of power for coal units closely align with reported variable costs (fig. B1) from the Merit Order Despatch of Electricity for Rejuvenation of Income and Transparency (MERIT), Ministry of Power database. This database reports variable generation costs based on long-term power purchase agreements (PPAs) between generators and distribution companies (Discoms). For lignite-based power plants, due to the lack of reliable lignite price data, I assume the reported variable cost from the MERIT India database.

Panel A compares the calculated variable cost of all power plants in the MERIT database for the year 2023 with their reported variable cost, yielding a moderately high correlation of 0.66.

In Panel B, I refine the analysis by eliminating three categories of plants: lignite-based plants, those receiving more than 90 percent of their coal supply from imports, and plants with MERIT variable costs lower than calculated variable costs. This refinement removes 37 out of 134 power plants from Panel A.

It is important to note that the MERIT database doesn't report the "true" variable cost faced by power plants that procure imported coal or buy it at e-auctions. Short-term changes in such fuel costs are not allowed pass-through in the power purchase agreements (PPAs). Consequently, during periods of high market prices for coal, power plants may become economically unviable to operate despite showing "low" variable costs in their PPAs. This discrepancy led to a complex situation. Until the April 11, 2017 Supreme Court judgment, these power plants would claim compensatory tariffs in addition to the PPA tariff to offset high fuel costs. This decision halted this practice, and tariff revisions are now subject to contract renegotiation to amend the PPAs.[21] Given these complexities and the potential mismatch between reported and actual costs, I disregard the MERIT variable cost for these power plants

[21] For instance, Mundra TPS signed a supplemental PPA with Haryana after months of renegotiation with a significantly higher tariff https://www.hindustantimes.com/cities/chandigarh-news/adani-power-agrees-to-sign-supplementary-ppa-with-haryana-101677691257073.html

in my analysis.

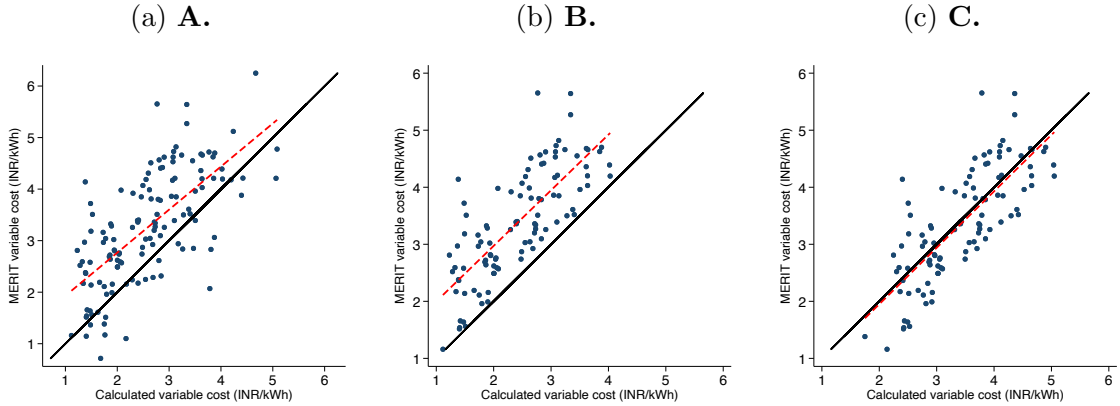
Among the remaining plants, only 7 power plants have calculated variable costs higher than MERIT variable costs, after excluding lignite-based plants and those receiving less than 95 percent of their coal supply domestically. This suggests that, according to MERIT figures, these 7 plants face lower fuel costs than linkage-based power plants. Such a scenario can occur, for example, due to changes in allocated mines towards the end of the study period, as part of ongoing coal cost rationalization efforts. [22]. For instance, the data shows that in 2022, towards the end of the study period, the Talwandi Sabo thermal power station in Punjab began sourcing part of its supply from Madhya Pradesh instead of Odisha, significantly reducing its transport costs.

Such re-allocations occurred late in the study period. Therefore, for these 7 power plants, I disregard the variable costs reported in MERIT and continue using the calculated variable costs in subsequent analyses.

After these adjustments, focusing on the remaining 97 power plants yields a very high correlation of 0.81, lending credibility to my calculated variable costs. Finally, as shown in Panel C, I adjust the calculated variable cost of power by 1.03 INR/kWh to close the gap between the lines of best fit and the 45-degree line in Panel B. This adjustment achieves parity between MERIT variable costs and calculated variable costs while preserving the merit order generated by the calculated variable costs. These adjusted variable costs are used throughout the paper for further analysis.

[22]See Ministry of Coal's press release for a great summary of these efforts: https://pib.gov.in/PressReleasePage.aspx?PRID=1953997#: :text=Four%20Rounds%20of%20Rationalization%20of,642

Appendix Figure B1. Comparison with MERIT database



Comparison of calculated variable cost of electricity to reported variable cost from Government of India MERIT database with lines of best fit (red dashed) and 45-degree line (black solid). Each point on the graph represents annual average monthly variable cost of electricity generation at one power plant. Panel (A) provides a comparison for all power plants in the MERIT database. Then, panel (B) eliminates lignite based plants, plants receiving greater 90 percent of their coal supply from imports and plants with MERIT variable costs lower than calculated variable costs. Panel (C) further adjusts the comparison by adding 1.03 INR/kWh to the calculated variable costs, closing the gap between line of best fit and 45-degree line in Panel (B).

B.1 Coal Price Construction

B.1.1 Basic Price of Coal

Price of linkage based coal: Power plants with linkage or fuel supply agreements (FSAs) with Coal India Limited (CIL) or Singareni Collieries Company Limited (SCCL) receive coal at administered prices. These prices are notified aperiodically by CIL and SCCL and are reported in rupees per tonne [23]. The methodology for determining the price of linkage based coal involves collecting grade-specific notified prices from both CIL and SCCL. Here, "grade" refers to the calorific value of coal, expressed in kilocalories (kcal) per kg. Each power plant is then assigned a "minemouth" (or "pithead") price based on the average grade of coal it consumes.

E-auction price: Some power plants source a portion of their coal supply from e-auctions conducted monthly by both CIL and SCCL. These e-auctions are conducted under different schemes, with power producers mainly bidding for coal under the spot e-auction and special forward e-auction schemes.

The Spot E-Auction scheme, operational since November 2007, allows any Indian

[23]The notified prices of coal are available on the websites of CIL and SCCL.

buyer to procure coal through a consumer-friendly single window for their own consumption or for trading. The Special Forward E-Auction scheme was introduced in 2015-16 to make coal available to power producers with flexible lifting periods.

To construct the e-auction price, I collect data on the average monthly price (reported as a percentage premium over notified price of coal) and the total quantity of coal auctioned by scheme.[24] The average e-auction price for power producers is then calculated as the weighted average of the prices from the two schemes, with weights based on the quantities procured under each. For months with missing data, I use the average scheme-specific price and quantity from the Ministry of Coal's Annual Report.[25] I assume this calculated price to be the "pithead" price for the share of coal procured from e-auctions.

Price of imported coal: There is no plant-specific price data for imported coal in India. To address this, I develop a methodology that leverages available data sources and makes informed assumptions about coal quality, measured in terms of Gross Calorific Value (GCV) and origins. My primary data source is the Directorate General of Commercial Intelligence and Statistics (DGCIS), Ministry of Commerce and Industry which reports monthly quantity and value of imported goods, categorized by country of origin and 8-digit product code. Power plants in this study use steam coal, corresponding to product code 27011920. Notably, approximately 90 percent of this coal is imported from two main sources: Indonesia and South Africa. The quality of coal from these sources differs significantly. I assume that Indonesian coal has an average GCV of 4600 kcal/kg, while South African coal has a higher average GCV of 6000 kcal/kg[26].

Using the DGCIS data, I first calculate the monthly average price of imported coal. I then estimate the average GCV of imported coal by taking a weighted average of the GCVs of Indonesian and South African coal. This calculation yields an average GCV of 4900 kcal/kg for imported coal. To validate this estimate, I compare it with the average GCV reported by power plants using imported coal, finding a close match that supports my approach.

[24] Data sourced from monthly reports to the Cabinet, available at: https://coal.gov.in/en/public-information/monthly-report-cabinet

[25] See the chapter "Coal Distribution and Marketing" in the Ministry of Coal's Annual Reports, available at: https://coal.gov.in/en/public-information/reports/annual-reports

[26] See the "Revised Standard Operating Procedure in respect of National Coal Index" published by the Ministry of Coal for the source behind these assumptions, available at: https://coal.nic.in/sites/default/files/2021-03/05-03-2021a-nci.pdf

Individual power plants vary in the GCV of the imported coal they use and consequently face different prices. To account for this variation, I next construct a plant-specific GCV price for imported coal. My approach involves expressing the plant-specific reported GCV as a convex combination of the GCVs of Indonesian and South African coal. The weights in this combination are based on the respective country shares in the total imported quantity. For example, if a plant reports using coal with a GCV of 5000 kcal/kg, this implies approximately 71.4 percent of the coal (by weight) came from Indonesia and 28.6 percent from South Africa. The price corresponding to that specific grade is then estimated as the weighted sum of the prices of Indonesian and South African coal, using these same weights.

To ensure the reliability of my estimates, I validate them using a subsample of grade-specific coal import prices sourced from CoalMint. I observe close alignment between my estimates and this independent data source, lending additional confidence in the methodology.

B.1.2 Taxes and surcharge

The basic coal price, as constructed in the previous section, is subject to a range of surcharges, government taxes, and levies. These additional costs vary depending on the source of the coal (linkage-based, e-auction, or imported) and the specific mining company involved.

For linkage-based coal, various surcharges are imposed by Coal India Limited (CIL) and Singareni Collieries Company Limited (SCCL) over the "pithead" price. These surcharges differ between the two companies. For CIL, these include Rs. 270 per tonne on pithead price of Run of Mine (ROM) coal for the supply of steam coal, sizing charges of Rs. 56 per tonne, loading charges of Rs 29 per tonne, and surface transportation charge for transportation of coal from pithead to loading points (assuming 0-3 KM distance) of Rs 40 per tonne. For SCCL, these include Rs. 220 per tonne for the supply of steam coal, sizing charges of Rs. 80 per tonne, loading charges of Rs 60 per tonne, and surface transportation charge of Rs 45 per tonne. SCCL imposes additional charges that vary over time, such as land adjustment fee, forest permit fee, fuel surcharge, explosive cost adjustment, and a charge towards Pension fund.[27].

[27] As of January 1, 2023, SCCL charges include a land adjustment fee of Rs. 61 per tonne, forest permit fee of Rs. 10 per tonne, fuel surcharge of Rs. 686 per tonne, explosive cost adjustment of Rs. 146.4 per tonne, and "Corpus of CMPS 1998 (Pension fund)" charge of Rs. 10 per tonne. See

I add these charges to the notified price and generate what constitutes as the basic price of coal. Additional government taxes and duties are then applied to this basic price. [28].

In my calculations, I include a royalty of 14 percent, with the exception of coal sourced from West Bengal, which charges a different royalty for coal sourced from its mines[29]. Furthermore, a District Mineral Foundation (DMF) Trust charge of 30 percent on the Royalty and a National Mineral Exploration Trust (NMET) charge of 2 percent on the Royalty are applied. The Goods and Services Tax (GST) is levied at 5 percent on the taxable value, and a GST Compensation Cess of Rs. 400 per tonne of dispatch is also imposed[30].

For coal procured through e-auctions, I apply the same set of surcharges, fees, and taxes on the "pithead" price realized in these auctions. A small subset of power plants source all or part of their coal from captive mines. For this supply, I assign the linkage based Coal India Limited price.

Imported coal is subject to a different tax structure. Unlike domestically sourced coal, imported coal is not subject to CIL/SCCL charges or Goods and Services Tax (GST). Instead, the government imposes a GST Compensation Cess of Rs. 400 per tonne and a 2.5 percent import duty on the imported price of coal, which I include in my calculations.

B.1.3 Distance calculation

Finally, to arrive at the plant-specific "delivered price" of coal, transportation charges must be added to the previously estimated coal price. The majority of power plants receive their coal via train, and these transportation charges, assessed in rupees per tonne, vary both over time and by the distance between the mine and the plant. I calculate these transportation charges as follows. First, railway freight rates are col-

details at: https://scclmines.com/scclnew/our-business_coal-prices.asp

[28]The additional charges imposed by Coal India Limited over and above the notified price are available at: https://bcclweb.in/S&M%20Documents/pricing767.pdf. The surface transportation charges are here https://www.secl-cil.in/writereaddata/stc%20notice(64dt18082022).pdf

[29]For the range of GCV values of coal supplied from West Bengal, the relevant royalty is Rs. 2.5 per tonne. See details at: https://coal.gov.in/sites/default/files/2019-11/180512_0_0.pdf

[30]The details of taxes, royalty, Goods and Service Tax (GST), Cess, and other levies imposed on coal are available at: https://sansad.in/getFile/annex/253/AU2088.pdf?source=pqars

lected from the Indian Railways website. These rates provide the basis for calculating the transportation costs.

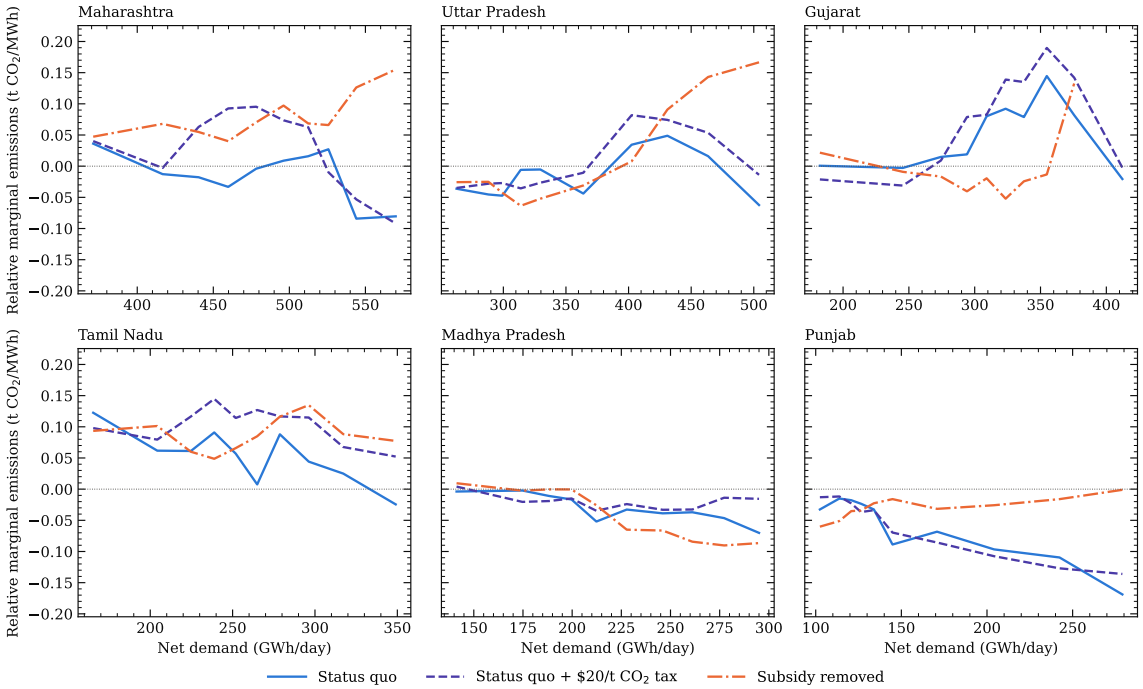
To calculate distance by rail between coal plants and coalfields, I combine plant geographic coordinates collected from The World Resources Institute with geospatial data on India's coalfields from the U.S. Geological Survey. These two datasets are then integrated with rail network data created by ML InfoMap to calculate the rail distance between each power plant and its coal source.[31]

The transportation cost per tonne of coal for each power plant is then calculated by multiplying the estimated distance with the relevant freight rate. This method provides a plant-specific transportation cost that accounts for both the distance traveled and the time-varying railway freight rates. A special case is made for "pithead" power plants, which are categorized as plants co-located next to a mine with a distance of less than 100 km from the mine. For these plants, the rates of the Merry-Go-Round (MGR) system of Indian Railways are applied. This system, designed for short-distance coal transport, typically offers more favorable rates for these nearby plants. Power plants designated for imported coal are assumed to incur zero surface transportation costs as these are typically located close to seaports.

[31] More information on these datasets can be found here: https://earthworks.stanford.edu/catalog/stanford-ww857qy4996

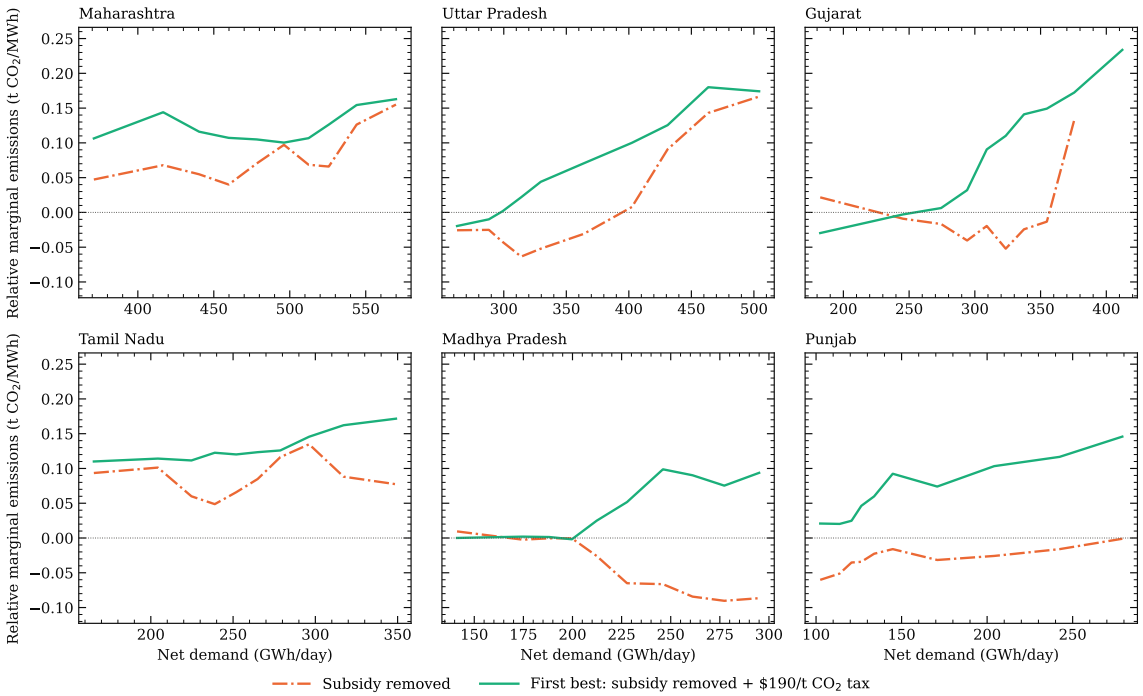
C Appendix Two

Appendix Figure C1. Marginal emissions curves by state: status quo, carbon tax and removal of the distortion



Marginal emissions curves for the six states with the highest net demand and private coal capacity. On each day, a state's marginal generator is the highest-cost generator supplying that state, whether under a long-term contract or not, with costs including any carbon tax; ties are weighted by generation. Emissions intensity is measured relative to the capacity-weighted fleet mean of coal and lignite generators (1.013 tonnes of CO_2 per MWh) and averaged within deciles of the state's own daily net demand. Panels share the vertical axis. Deciles whose average departs by more than 0.1 tonnes of CO_2 per MWh from the rolling median of the decile and its neighbors are omitted; this removes one point, the highest demand decile in Gujarat with the distortion removed (0.38 tonnes above the fleet mean), where the marginal generator is a lignite plant on most days. July 2021–December 2023.

Appendix Figure C2. Marginal emissions curves by state: removal of the distortion and the first best



Marginal emissions curves for the six states in Figure C1, comparing removal of the distortion with the first best, which removes the distortion and adds a carbon tax of $190 per tonne, the social cost of carbon. Definitions, sample and the omitted point for Gujarat are as in Figure C1.